Qin Jiushao, Shuxue Jiuzhang, fascicles 1-9

Chinese Original

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数书九章

Mathematical Treatise in Nine Sections

Fascicle I

The Dayan Category (大衍類)

By

Qin Jiushao (秦九韶)

c. 1202–1261 CE

Presented in the 7th year of Chunyou (淳祐七年), 1247 CE

One of the greatest masterpieces of Chinese mathematics,
containing the general Chinese Remainder Theorem (Dayan method),
numerical solutions of polynomial equations, surveying problems,
and commercial and financial mathematics.

Typeset in LaTeX with XeTeX

Contents

1 Introduction

The Shuxue Jiuzhang (数书九章, Mathematical Treatise in Nine Sections) is the magnum opus of the Song dynasty mathematician Qin Jiushao (秦九韶, c. 1202–1261). Completed in 1247 CE during the Chunyou era (淳祐七年), it stands as one of the greatest achievements in the history of Chinese mathematics and indeed of world mathematics.

The work is organized into nine categories (类), each containing nine problems (问), for a total of eighty-one problems. The nine categories are:

[label=1.]

1. **Dayan** (大衍) – The Great Derivation: Indeterminate analysis and the Chinese Remainder Theorem
2. **Tianshi** (天时) – Astronomical phenomena and calendar calculations
3. **Tianyu** (田域) – Land measurement and area calculation
4. **Cewang** (测望) – Surveying and observation
5. **Fuyi** (赋役) – Taxation and labor service
6. **Qian'gu** (钱谷) – Currency and grain commerce
7. **Yingjian** (营建) – Construction and engineering
8. **Junlv** (军旅) – Military affairs
9. **Shiyi** (市易) – Market transactions

Each problem follows a rigorous four-part structure:

- **Wen** (问) – The problem statement
- **Da** (答) – The numerical answer
- **Shu** (术) – The general method and algorithm
- **Cao** (草) – The detailed working (step-by-step calculation)

1.1 The Dayan Category (大衍类)

Fascicle I covers the first category, Dayan (大衍, "Great Derivation"), devoted to the general solution of systems of linear congruences—what is now known worldwide as the **Chinese Remainder Theorem**. Qin Jiushao's Dayan zongshu shu (大衍总数术, "Dayan General Method") provides a systematic algorithm, and his Dayan qiu yi shu (大衍求一术, "Dayan Method for Finding Unity") gives the procedure for finding modular multiplicative inverses.

The nine problems of the Dayan category are:

[label=1.]

1. **Shigua fawei** (蓍卦发微) – Divination by Yarrow Stalks
2. **Guli huiji** (古历会积) – Ancient Calendar Accumulation
3. **Tuiji tugong** (推计土功) – Estimating Earthworks
4. **Tuiku e qian** (推库额钱) – Calculating Treasury Funds
5. **Fentiao tuiyuan** (分粜推原) – Grain Distribution

6. **Chengxing jidi** (程行计地) – Journey Distance Calculation
7. **Chengxing xiangji** (程行相及) – Catching Up on a Journey
8. **Jichi xunyuan** (积尺寻源) – Finding Dimensions from Volume
9. **Yumi tuishu** (余米推数) – Calculating Rice Remainders

2 Preface (序)

周教六艺，数实成之。学士大夫所从事尚矣。其用本太虚生一，而周流无穷。大则可以通神明、顺性命，
若昔推策以迎日，定律而和气。髀距濬川，土圭度晷。天地之大，囿焉而不能外，况其间总总者乎？爰自
汉去古未远，有张苍、许商、马延年、耿寿昌、郑玄、张衡、刘洪之伦，或明天道而法传于后，或计功策
呜呼！乐有制氏，仅记铿锵之声，而谓与天地同和者止于是，可乎？今数术之书尚三十余家。天象历度谓
且天下之事多矣。古之人先事而计，计定而行。仰观俯察，人谋鬼谋，无所不用其谨，是以不愆于成。载
九韶愚陋，不闲于艺。然早岁侍亲中都，因得访习于太史，又尝从隐君子受数学。时际兵难，历岁遥塞，
时淳祐七年九月，鲁郡秦九韶叙。

2.1 The Nine Categories in Verse

昆仑旁薄，道本虚一。圣有大衍，微寓于《易》。奇余取策，群数皆捐。衍而究之，探隐之原。

数术之传，以实为体。其书九章，惟兹弗纪。历家虽用，用而不知。小试经世，姑推所为。

述大衍第一

七精回穹，人事之纪。追缀而求，宵星昼晷。历久则疏，惟智能革。不寻天道，模袭何益。

述天时第二

魁隗粒民，甄度四海。苍姬井之，仁政攸在。代远庶蕃，垦菑日广。步度庀赋，版图是掌。方圆异状，

述田域第三

莫高匪山，莫濬匪川。神禹奠之，积矩攸传。智创巧述，重差夕桀。求之既详，揆之罔越。崇深广远，

述测望第四

邦国之赋，以待百事。畦田经入，取之有度。未免力役，先商厥功。以衰以率，逸乃同功。汉犹近古，

述赋役第五

物等敛赋，式时府庾。粒粟寸丝，褐夫红女。商征边籴，后世多端。立缘为欺，上下俱殚。我闻理财，

述钱谷第六

斯城斯池，乃栋乃宇。宅坐寄命，以保以聚。鸠功雉制，竹个本章。匪究匪度，财蠹力伤。围蔡而裁，

述营建第七

天生五材，兵去未可。不教而战，维上之过。堂堂之阵，鹅鹳为行。营应规矩，其将莫当。师中之吉，

述军旅第八

日中而市，万民所资。贾贸墆鬻，利析铢锱。蹛财役贫，封君低首。豕末兼并，非国之厚。

述市易第九

3 The Dayan General Method (大衍总数术)

3.1 Editor’s Commentary on the Dayan Method

按大衍术，以各分数之奇零求各分数之总数。大而天行，小而物数，皆可御之。其法有求元、求定、求衍、
有求彼此不能度尽之诸数者，元数、定数是也。有求诸数皆能度尽之一数者，衍母数是也。有求诸数皆能度

3.2 The Four Types of Numbers

大衍总数术曰：置诸问数，类名有四。

[label=2.]

1. **元数** – 谓尾位见单零者。本门揲蓍、酒息、斛粜、砌砖、失米之类是也。
2. **收数** – 谓尾位见分厘者。假令冬至三百六十五日二十五刻，欲与甲子六十日为一会而求积日之类。
3. **通数** – 谓诸数各有分子分母者。本门问一会积年是也。
4. **复数** – 谓尾位见十或百及千以上者。本门筑堤并急足之类是也。

3.3 Finding the Definitive Numbers (求定数)

元数者：先以两两连环求等，约奇弗约偶。或约得五而彼有十，乃约偶弗约奇。或元数俱偶，约毕可存一位见
收数者：乃命尾位分厘作单零，以进所问之数，定位讫，用元数格入之。或如意立数为母，收进分厘以从所问
通数者：置问数通分纳子，互乘之，皆曰通数。求总等，不约一位，约众位，得各原法数，用元数格入之。或
复数者：问数尾位见十以上者，以诸数求总等，存一位约众位，始得元数。两两连环求等，约奇弗约偶，复乘

3.4 Computing the Derivative Numbers

求定数勿使两位见偶，勿使见一太多。见一多则借用繁，不欲借则任得一。
以定相乘为衍母，以各定约衍母，得各衍数。或列各定数于右行，各立天元一为子于左行，以母互乘子，亦得
Then each derivative number is reduced modulo its corresponding definitive number; the remainder is called the **odd number** (奇数). Using the Dayan Method for Finding Unity (大衍求一术), one finds the **multiplier** (乘率) for each. Multiplying each multiplier by its derivative number gives the **use number** (用数). The final answer is obtained by multiplying each remainder by its use number, summing, and reducing modulo the derivative mother (衍母).

3.5 The Dayan Method for Finding Unity (大衍求一术)

大衍求一术云：置奇右上，定居右下。立天元一于左上。先以右行上下两位，以少除多，所得商数，乃递互

In modern terms: Given coprime integers a (the “odd number”) and m (the “definitive number”), find k such that:

$$k \cdot a \equiv 1 \pmod{m}$$

This is the modular multiplicative inverse of a modulo m, computed by the Euclidean algorithm with back-substitution.

4 Problem 1: Shigua Fawei (蓍卦发微)

蓍卦发微 – Divination by Yarrow Stalks

问曰 《易》曰：大衍之数五十，其用四十有九。又曰：分而为二以象两，挂一以象三，揲之以四以象四时。三

The Book of Changes says: “The number of the Great Derivation is fifty; of these, forty-nine are used.” It also says: “Divide into two to symbolize the Two [heaven and earth]; hang one to symbolize the Three [heaven, earth, and man]; count off by fours to symbolize the Four Seasons.” Three changes produce a line; eighteen changes produce a hexagram. The problem asks to find the computational method of this derivation and the numbers involved.

答曰 衍母十二，衍法三。

一元衍数二十四，二元衍数一十二，三元衍数八，四元衍数六。以上四位衍数，计五十一。

一揲用数一十二，二揲用数二十四，三揲用数四，四揲用数九。以上四位用数，计四十九。

术曰 置诸元数，两两连环求等，约奇弗约偶。遍约毕，乃变元数皆曰定母，列右行。各立天元一为子，列左行

草曰 置一、二、三、四列右行，立天元一列左行。

以右行一、二、三、四互乘左行：一乘一得一，二乘一得二，三乘二得六，四乘六得二十四。各得衍数。

次以定母满法去衍数：

- 以十二除二十四得二余零，一元衍数为二十四
- 以十二除一十二得一余零，二元衍数为一十二
- 以八除八得一余零，三元衍数为八
- 以六除六得一余零，四元衍数为六

各满定母去之，得奇数。以大衍求一入之，得乘数：

- 得二十三为乘率
- 得五十一为乘率
- 得七为乘率

以乘率各乘衍数，得用数：

- 一揲用数一十二
- 二揲用数二十四
- 三揲用数四
- 四揲用数九

以上四位用数，计四十九。

This problem applies the Dayan method to the classical yarrow-stalk divination procedure of the Book of Changes (I Ching), interpreting the numerological operations in terms of congruence arithmetic.

5 Problem 2: Guli Huiji (古历会积)

古历会积 – Ancient Calendar Accumulation

问曰 问古历会积，欲求积年。以历法积元为问，历过无考，但据近世演纪之法，推而上之。

This problem concerns the calculation of the accumulated years (积年) in ancient calendar systems. Given certain astronomical periodicities, one must find a number that simultaneously satisfies multiple congruence conditions.

答曰 答曰：积年若干。

术曰 术曰：以大衍求之。置元历相关诸周期，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍

This problem applies the Dayan method to astronomical calendar calculations, finding a common epoch that aligns multiple celestial cycles (such as the tropical year, synodic month, and sexagenary cycle). This was one of the primary applications of the Dayan method in traditional Chinese astronomy.

6 Problem 3: Tuiji Tugong (推计土功)

推计土功 – Estimating Earthworks

问曰 问：四县（甲、乙、丙、丁）共同筑堤，根据各县的物力分配筑堤任务。

- 甲县物力：138,600贯
- 乙县物力：146,300贯
- 丙县物力：192,500贯
- 丁县物力：184,800贯

筑堤标准：每770贯物力分配一名劳动力，每人每天筑堤60平方尺。

进度情况：甲县和乙县已完成任务，丙县剩余51丈，丁县剩余18丈。

答曰 答曰：堤的总长度为19里235步5尺。

各县所筑长度：

- 甲县：1,260丈
- 乙县：1,768步5尺6寸
- 丙县：4里328步5尺6寸
- 丁县：与前三县相应

术曰 术曰：以大衍求之。置各县物力，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍

草曰 计算各县每日筑堤长度：

- 甲县：54丈
- 乙县：57丈
- 丙县：75丈
- 丁县：72丈

使用大衍求一术，计算各县的复数和衍数：

- 甲县复数：27，乙县复数：19
- 丙县复数：25，丁县复数：8

计算衍母和衍数：

- 衍母：102,600
- 甲县衍数：3,800
- 乙县衍数：5,400
- 丙县衍数：4,140
- 丁县衍数：12,825

计算各县的乘率：

- 甲县乘率：23
- 乙县乘率：5
- 丙县乘率：19
- 丁县乘率：1

计算各县的用数：

- 甲县用数：17,400
- 乙县用数：27,000
- 丙县用数：77,976
- 丁县用数：12,825

计算总长度：

- 丙县剩余长度：3,976,776丈
- 丁县剩余长度：238,500丈
- 总剩余长度：4,215,276丈

堤的总长度：19里235步5尺。

7 Problem 4: Tuiku E Qian (推库额钱)

推库额钱 – Calculating Treasury Funds

问曰 问有外邑七库，日纳息足钱适等，递年成贯整纳。近缘见钱希少，听各库照当处市陌准解旧会。其甲库有

（七库依次为甲、乙、丙、丁、戊、己、庚，市陌分别为12, 11, 10, 9, 8, 7, 6文。甲库余10文，乙丙无余，丁

答曰 答曰：诸库纳日息足钱二十贯九百五十文。

展省三十五贯文。

各库旧会：

- 甲库日息旧会：二百二十四贯五百一十文
- 乙库日息旧会：二百四十五贯文

- 丙库日息旧会：二百六十九贯五百文
- 庚库日息旧会：四百四十九贯一百四文

术曰 术曰：以大衍求之。置甲库市陌，以递减数减之，各得诸库原陌。连环求等，约奇弗约偶，得定母。诸定
草曰 置甲库市陌一十二，递减一得一十一为乙库陌，十为丙库陌，九为丁库陌，八为戊库陌，七为己库陌，六
连环求等约讫：

- 甲得一
- 乙得十一
- 丙得五
- 丁得九
- 戊得八
- 己得七
- 庚得一

各为定母。
先以诸定相乘，得27,720为衍母。 次以诸定互乘诸子：

- 甲得27,720
- 乙得2,520
- 丙得5,544
- 丁得3,080
- 戊得3,465
- 己得3,960
- 庚得27,720

各为衍数。

8 Problem 5: Fentiao Tuiyuan (分粜推原)

分粜推原 – Grain Distribution

问曰 有兄弟三人，各将收获之米送往不同的地方：

- 甲将米送往官场，余3斗2升
- 乙将米送往安吉乡，余7斗
- 丙将米送往平江，余3斗

官斛每石八斗三升，安吉斛每石一石一斗，平江斛每石一石三斗五升。欲求三人共收获米数以及各自分米数几何
答曰 答曰：三人共收获米738石。
各分米数：

- 甲：296石，余3斗2升

- 乙：223石，余7斗
- 丙：182石，余3斗

术曰 术曰：以大衍求之。置各斛率（单位：升）：官斛83升，安吉斛110升，平江斛135升。连环求等，约奇
草曰 置官斛83升、安吉斛110升、平江斛135升，连环求等。
求总等：得总等1，不约。连环求等，得定母：246,510。
各斛衍数：

- 官斛衍数：2,970
- 安吉衍数：2,241
- 平江衍数：1,826

以大衍求一术，得乘率：

- 官斛乘率：51
- 安吉乘率：7
- 平江乘率：23

以乘率乘衍数得用数：

- 官斛用数：$2{,}970 \times 51 = 151{,}470$
- 安吉用数：$2{,}241 \times 7 = 15{,}687$
- 平江用数：$1{,}826 \times 23 = 41{,}998$

次置各余数乘用数：

- 甲余32升 $\times$ 官斛用数
- 乙余70升 $\times$ 安吉用数
- 丙余30升 $\times$ 平江用数

并为总数，满衍母246,510去之，不满为所求率数。
展算得：分米246石，以兄弟三人因之，得738石为共米。
置分米246石，各以官斛8斗3升、安吉斛1石1斗、平江斛1石3斗5升约之：

- 甲得296石，余3斗2升
- 乙得223石，余7斗
- 丙得182石，余3斗

各为粜过及余米，合问。

9 Problem 6: Chengxing Jidi (程行计地)

程行计地 – Journey Distance Calculation

问曰 军师派遣三名急足（甲、乙、丙）分别前往都下报捷。

- 甲每天行300里
- 乙每天行240里
- 丙每天行180里

甲提前数日申未到，乙后数日未正到，丙今日辰末到。欲知从军前到都下的距离以及三人各自行走的天数几何?

答曰 答曰：从军前到都下的距离为3,300里。

- 甲行11天
- 乙行13天4时
- 丙行18天2时

术曰 术曰：以大衍求之。置三人日行里数，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母

草曰 置甲300里、乙240里、丙180里，求总等，连环求等。

计算得乘率：

- 甲乘率：4
- 乙乘率：1
- 丙乘率：7

通过乘率乘以各自的用数，得到总用数，再通过总用数除以衍母，得到最终结果。

验证：

- $3{,}300 \div 300 = 11$天
- $3{,}300 \div 240 = 13$天余120里 = 13天4时（1天=240里，120里=半天=4时）
- $3{,}300 \div 180 = 18$天余60里 = 18天2时

合问。

10 Problem 7: Chengxiang Xiangji (程行相及)

程行相及 – Catching Up on a Journey

问曰 问：甲、乙、丙三人行程相关。甲先行若干日，乙后行，丙又后行。各以不同速度行进。求某处三人相遇

This problem involves multiple travelers with different speeds and starting times, requiring the use of the Dayan method to solve a system of congruences for finding meeting times and distances.

答曰 答曰：所求距离及相关数值若干。

术曰 术曰：以大衍求之。置各人行程速度及时间差，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各

11 Problem 8: Jichi Xunyuan (积尺寻源)

积尺寻源 – Finding Dimensions from Volume

问曰 问欲砌基一段，见管大小方砖六门城砖四色。令匠取便，或平或侧，只用一色砖砌，须要适足。匠以砖量

- 大方：广多六寸，深少六寸
- 小方：广多二寸，深少三寸
- 城砖：长广多三寸，深少一寸
- 六门砖：长广多三寸，深多一寸

皆不合匝，未免修破砖料裨补。其四色砖尺寸：

- 大方：方一尺三寸
- 小方：方一尺一寸
- 城砖：长一尺二寸，阔六寸，厚二寸五分
- 六门砖：长一尺，阔五寸，厚二寸

欲知基深广几何?

答曰 答曰：深三丈七尺一寸，广一丈二尺三寸。

This is a classic problem of finding a common multiple with given remainders. The dimensions of the foundation must be such that when measured with each type of brick (in different orientations), there are specific remainders. The Dayan method is used to find such dimensions.

术曰 术曰：以大衍求之。置四色砖尺寸及所余尺寸，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各

12 Problem 9: Yumi Tuishu (余米推数)

余米推数 – Calculating Rice Remainders

问曰 问：有米若干，以不同容量的容器量之，各余若干。欲求米之总数。

This problem involves finding the total number of units of rice when measured with containers of different capacities, leaving different remainders each time. It is the classic “unknown quantity” problem that leads directly to the Chinese Remainder Theorem.

答曰 答曰：所求米总数若干。

术曰 术曰：以大衍求之。置各容器容量为问数，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约

This problem is structurally identical to the famous “problem of the unknown quantity” from the Sunzi Suanjing (孙子算经), which asks: “We have things of which we do not know the number; if we count them by threes, the remainder is 2; if by fives, the remainder is 3; if by sevens, the remainder is 2. How many things are there?” Qin Jiushao’s general method solves all problems of this type systematically.

13 Mathematical Analysis of the Dayan Method

13.1 The Chinese Remainder Theorem

The Dayan method solves systems of simultaneous linear congruences. Given a set of pairwise coprime moduli $m_1, m_2, \ldots, m_n$ and remainders $r_1, r_2, \ldots, r_n$, find N such that:

$$\begin{aligned} N &\equiv r_1 \pmod{m_1} \\ N &\equiv r_2 \pmod{m_2} \\ &\vdots \\ N &\equiv r_n \pmod{m_n} \end{aligned}$$

13.2 Qin Jiushao’s Algorithm

Step 1: Definitive Numbers (求定数). Given the original numbers (元数), reduce them pairwise using the greatest common divisor, keeping them coprime. The rule “约奇弗约偶” (reduce the odd one, not the even one) ensures the product will have unique factorization.
Step 2: Derivative Mother (求衍母). Compute $M = m_1 \times m_2 \times \cdots \times m_n$.
Step 3: Derivative Numbers (求衍数). Compute $M_i = M/m_i$ for each i.
Step 4: Odd Numbers (求奇数). Compute $g_i = M_i \bmod m_i$.
Step 5: Finding Unity (大衍求一术). For each i, find k_i such that:

$$k_i \cdot g_i \equiv 1 \pmod{m_i}$$

This is the modular multiplicative inverse.
Step 6: Use Numbers (求用数). Compute $u_i = k_i \cdot M_i$.
Step 7: Final Computation. The solution is:

$$N = \sum_{i=1}^{n} r_i \cdot u_i \pmod{M}$$

13.3 Historical Significance

Qin Jiushao’s method predated similar European work by several centuries. The algorithm for finding modular inverses (大衍求一术) is essentially the extended Euclidean algorithm. The method was later applied by Guo Shoujing (郭守敬) to astronomy, by Li Ye (李冶) to geometry, and was transmitted to Europe where it became known as the “Chinese Remainder Theorem.”

Gauss gave the first complete treatment in Europe in his Disquisitiones Arithmeticae (1801), calling it the “problem of finding a number that, when divided by given numbers, leaves given remainders.” The theorem is now recognized as one of the fundamental results in number theory.

About This Edition

This typeset edition presents Fascicle I of Qin Jiushao’s Shuxue Jiuzhang (Mathematical Treatise in Nine Sections), covering the Dayan category (大衍类) with its nine problems on the Chinese Remainder Theorem and related indeterminate analysis.

The source text is based on the Siku Quanshu (四库全书) edition as preserved in the Chinese Textual Tradition, with cross-references to the Yijia Tang (宜稼堂) edition. The editorial commentary (按) included in the Siku edition has been preserved where available.

Technical Notes

- Typeset with XeTeX
- Chinese font: Noto Sans CJK SC
- The traditional structure of 问 (question), 答 (answer), 术 (method), and 草 (working) has been preserved

Further Reading

- Libbrecht, U. Chinese Mathematics in the Thirteenth Century. MIT Press, 1973.
- Qian, Baocong. Zhongguo Shuxue Shi (History of Chinese Mathematics). Science Press, 1964.
- Martzloff, J.-C. A History of Chinese Mathematics. Springer, 1997.

Mathematical Treatise in Nine Sections

Shuxue Jiuzhang – 數學九章

Fascicle 2

天時類

Tian Shi Lei

(Heavenly Patterns: Calendar, Astronomy and Weather)

Qin Jiushao

秦九韶 (1202–1261)

Song Dynasty, Chunyou 7th year (1247)

Presented in modern typesetting with English commentary
based on the Siku Quanshu (Complete Library of the Four Treasuries) edition

Edited and typeset from the original Chinese mathematical text

Contents

Introduction to Fascicle 2

The **Tian Shi Lei** (天時類, "Heavenly Patterns") is the second of nine chapters in Qin Jiushao's *Shuxue Jiuzhang* (數學九章, Mathematical Treatise in Nine Sections). This fascicle contains **nine problems** divided into two parts, dealing with:

- Calendar computation and astronomical predictions (Problems 1–5)
- Measurement of rainfall and snowfall (Problems 6–9)

Qin Jiushao's preface to the *Shuxue Jiuzhang* (1247) explains his motivation:

> *"The art of numbers, rooted in the Great Void, generates One and circulates without end. In its large aspect it can penetrate the divine and comply with destiny; in its small aspect it can govern worldly affairs and classify the myriad things. How can it be viewed as shallow or narrow?"*

The nine problems of this fascicle demonstrate the practical application of mathematical techniques to calendar reform, planetary motion, and meteorological measurement—topics of vital importance to agricultural society.

1 Volume 2, Part 1 (卷二上)

1.1 Problem 1: Calculating Qi and Regulating the Calendar

推氣治歷 (Tui Qi Zhi Li)

問 The Imperial Astronomer observed the heavenly way:

- In the 4th year of Qingyuan (慶元四年, 1198, year Wu-wu 戊午), the winter solstice was at 39 days, 92 *ke* (刻), 45 *fen* (分).
- In the 3rd year of Shaoding (紹定三年, 1230, year Geng-yin 庚寅), the winter solstice was at 32 days, 94 *ke*, 12 *fen*.

It is desired to find, for the intermediate year Jiatai Jiazi (嘉泰甲子, 1204): the *qi bone* (氣骨), the *year remainder* (歲餘), and the *dou fraction* (斗分).

按 The Shaoding 3rd year Geng-yin winter solstice actually marks the beginning of Shaoding 4th year Xin-mao (辛卯). Xin-mao is 34 years from Wu-wu, giving 33 accumulated years.

答曰

- Qi bone: 11 days, 38 *ke*, 20 *fen*, 81 *miao* (杪), 80 small fractions.
- Year remainder: 5 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.
- Dou fraction: 0 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.

術曰曰 First take the number of years between the two observations as the divisor. Place the earlier observation's days, *ke*, and *fen*, and subtract from the later observation's days, *ke*, and *fen*; the remainder is the rate. (If insufficient for subtraction, add the ji-ce 紀策). Accumulate by adding ji-ce repeatedly until the result is suitable for the heavenly way—using a number above 5 days as the dividend. Divide the dividend by the divisor to obtain the year remainder. Remove whole days; the remainder is the dou fraction. Multiply the year remainder by the number of years from the earlier observation to the desired intermediate year, add to the earlier observation's days, *ke*, and *fen*, and remove full ji-ce; the remainder is the desired qi bone.

按 (Editor's note:) The qi bone is the days and fractions from the winter solstice to the Jiazi day after midnight. The year remainder is the residue when the solar year length is reduced by six Jiazi cycles. The dou fraction is the residue when the solar year is reduced by 365 days. This method does not yet know the precise solar year length, hence it uses the difference between two observed qi bones as the dividend and accumulated years as divisor.

1.2 Problem 2: Regulating the Calendar and Intercalary Months

治歷推閏 (Zhi Li Tui Run)

問 For the Kaixi calendar (開禧曆), taking the 4th year of Jiatai (嘉泰四年, 1204, year Jiazi 甲子):

- The celestial winter solstice is at 11 days, day-token Yi-hai (乙亥), 44 *ke*, 61 *fen*, 54 *miao*.
- The 11th-month new moon (經朔) is at 1 day, day-token Yi-chou (乙丑), 75 *ke*, 55 *fen*, 62 *miao*.

Question: What are the *run bone* (閏骨) and *run rate* (閏率)?

答曰

- Run bone: 9 days, 69 *ke*, 5 *fen*, 91 *miao*, with remainder 121/(169 fractions).
- Run bone rate: 163,771.

術曰曰 Take the day-factor (日法) to convert the qi and new-moon days, *ke*, *fen*, and *miao* into qi-bone and new-moon-bone fractions respectively. The qi-bone fraction, when reduced by the approximation rate (約率), if exactly divisible, is usable; or if the remainder after rounding is below one *ke*, it is also usable. Then subtract from the new-moon-bone fraction; the remainder is the run bone rate. Divide by the day-factor to obtain the run bone *ce* (策).

按 (Editor's note:) If one directly subtracts the new moon fraction from the winter solstice fraction, the remainder is 9 days, 69 *ke*, 5 *fen*, 92 *miao*, giving the run bone *ce*. The original working only differs by 48/(169 fractions). The reason for not directly subtracting but instead using common denominators, approximation, and repeated multiplication/division is to facilitate subsequent calculations.

1.3 Problem 3: Deriving Calendar Epoch Calculations

治歷演紀 (Zhi Li Yan Ji)

問 The Kaixi calendar has an accumulated year count of 7,848,183. It is desired to know the derivation of this computation: adjusting the day-factor, finding the new-moon remainder, new-moon rate, dou fraction, year rate, year intercalation, entry-year, entry-intercalation, new-moon fixed bone, intercalation floating bone, intercalation shrinkage, ji rate, qi yuan rate, yuan intercalation, yuan number, qi equal-rate, cause-rate, *bu* rate, new-moon equal-number, cause-number, *bu* number, and new-moon accumulated year—**23 items** in total. What is each?

答曰 (Selected key results:)

- Day-factor (日法): 16,900
- New-moon remainder (朔餘): 8,967
- New-moon rate (朔率): 499,067
- Dou fraction (斗分): equivalent to 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions
- Year rate (歲率): 6,172,608
- Year intercalation (歲閏): 20,728
- Entry-year (入元歲): 9,240
- Entry-intercalation (入閏): 31,068

術曰曰 Use the method of Dayan (大衍術) to find the qi fraction, new-moon fraction, ji fraction, and derived mother (衍母), and all the proper use-numbers (正用). Divide the derived mother by the qi fraction to obtain the accumulated years; by the new-moon fraction to obtain the accumulated months; by the ji fraction to obtain the accumulated ji. Multiply by 60 to obtain accumulated days. Multiply the qi bone by the ji proper-use to obtain the ji total; multiply the run bone by the new-moon proper-use to obtain the new-moon total. Combine the two totals, remove full derived mothers, and the remainder is the sought rate-dividend. Divide by the qi fraction to obtain the calendar-passed (歷過) years; subtract from the accumulated years to obtain the years not yet reached.

This problem demonstrates the full power of Qin Jiushao's **Dayan method** (大衍求一術) for solving systems of congruences, applied here to calendar computation. The 23 quantities computed form a complete system for deriving the epoch parameters of the Kaixi calendar from astronomical observations.

1.4 Problem 4: Calculating Planetary Motion

綴術推星 (Zhui Shu Tui Xing)

問 The year-star (Jupiter, 歲星) in conjunction-hidden phase (合伏段): after 16 days 90 *fen*, it moves 3 degrees 90 *fen*, removing 13 degrees from the sun before becoming visible. Then it proceeds directly (順行) for 113 days, 17 degrees 83 *fen*, before halting (乃留). It is desired to know:

- The conjunction-hidden segment (合伏段)
- The morning-rapid initial segment (晨疾初段)
- The standard degree (常度)
- The initial motion-rate (初行率)
- The final motion-rate (末行率)
- The mean motion-rate (平行率)

What is each?

術曰曰 (Method of *Zhui Shu*):

1. For the conjunction-hidden segment: The constant degree is the total number of days. The initial motion rate is the total degrees moved. The final motion rate equals the initial rate minus the daily difference (日差) multiplied by (days -1).
2. For the morning-rapid initial segment: The constant degree is the given number of days. The initial rate is the final rate of the previous segment. The final rate equals the initial rate minus the daily difference multiplied by (days -1).
3. For the mean motion rate: Add the initial and final rates and halve; this gives the parallel (mean) rate.

按 (Editor's note:) The five planets' motions involve non-uniform acceleration and deceleration. The method here takes only the middle values of the conjunction-hidden and direct-visibility segments, then uses successive addition/subtraction to derive the daily motion. The ancient method's approximation is particularly evident for the five planets. The original text's language was mostly hidden and obscure; the following detailed explanation reveals the comparison between ancient and modern precision.

Key calculations:

- Conjunction-hidden segment:
  - Constant degree: 16 days 90 *fen*
  - Initial rate: 22 *fen* 7 *miao*
  - Final rate: 21 *fen* 96 *miao*
  - Mean rate: approximately 22 *fen* (after rounding)
- Morning-rapid initial segment:
  - Constant degree: 30 days
  - Initial rate: 21 *fen* 96 *miao*
  - Daily difference: 11 *miao* 23 small-*fen* 66 small-*miao*

1.5 Problem 5: Measuring the Sun's Shadow

揆日究微 (Kui Ri Jiu Wei)

問 Among all historical shadow-measurements, only the Tang dynasty's Dayan calendar (大衍曆) was the most precise. This dynasty's Chongtian calendar (崇天曆) gives for Yangcheng (陽城):

- Winter solstice shadow: 1 *zhang* 2 *chi* 7 *cun* 1 *fen* 50 *miao*
- Summer solstice shadow: 1 *chi* 4 *cun* 7 *fen* 70 *miao*

These agree with the Dayan calendar. Now the Kaixi calendar (開禧曆) gives for Lin'an (臨安府):

- Winter solstice shadow: 1 *zhang* 8 *cun* 2 *fen* 25 *miao*
- Summer solstice shadow: 9 *cun* 1 *fen*

It is desired to find: after how many days past the summer solstice at Lin'an will the shadow equal the Yangcheng summer solstice shadow? And comparing with the Dayan calendar's shadow, by how much do they differ?

答曰 Five days after the Great Heat (大暑後五日) at noon; the shadow length is 1 *chi* 4 *cun* 8 *fen* 85 *miao* (half). The shadow difference is 6 *cun* 1 *fen* 29 *miao* (less).

術曰曰 Use the multiplication-rate method, with the section-rate (節率) entering into it.

1. Place Lin'an's measured winter solstice shadow, subtract the summer solstice shadow; the remainder is the shadow-difference (景差), taken as dividend.
2. Place the 象限度 (91 degrees 31 *fen* 44 *miao*), add 11 degrees 25 *fen* 27.5 small-*fen*; treating degrees as *cun*, obtain 102 *cun* 5671.5 *fen* 50 *miao* as divisor.
3. Divide the dividend by the divisor, obtaining 0.9664...; discard the remainder and square to obtain the section floating-number (節泛數): 9,340 *fen*.
4. Multiply by the section multiplication-rates for each solar term, add the summer solstice shadow-squared, take the square root to obtain each solar term's shadow length.

按 (Editor's note:) Each solar term's shadow length was determined by actual measurement at the time and does not need computation. The method presented here deliberately creates obscurity to puzzle the reader. Detailed examination shows: the 象限度 plus 11+ degrees is taken as divisor; the shadow-difference divided by this and then squared gives the section-rate. Each solar term has a multiplication-rate; multiplying the section-rate by this and adding the summer solstice shadow-power gives each term's shadow-power. Thus the section-rate is an artificially extracted number.

2 Volume 2, Part 2 (卷二下)

2.1 Problem 6: Measuring Rain with the Heavenly Pool Basin

天池測雨 (Tian Chi Ce Yu)

問 Nowadays every prefecture and district has a "heavenly pool" basin (天池盆) for measuring rainwater. But people only know that the water in the basin equals the rain received, and do not understand that different vessel shapes receive different amounts of rain—so one cannot directly take the basin measurement as the flat-ground rain amount.

Suppose the basin has:

- Mouth diameter (口徑): 2 *chi* 8 *cun* (28 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)
- Depth (深): 1 *chi* 8 *cun* (18 *cun*)
- Rainwater depth inside (接雨水深): 9 *cun*

What is the equivalent flat-ground rain depth?

答曰 Flat-ground rain depth: **3 *cun***.

術曰曰 Use the method for circular cones (圓錐術):

1. Multiply the mouth diameter by the base diameter, and also square each; combine these three values. Multiply by half the depth; then multiply by 11 and divide by 7 to obtain the lower volume (下積).
2. Take half the depth plus the rain depth, subtract from the total depth to obtain the upper depth (上深).
3. Subtract the base diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth, add the result to the mouth-diameter times half-depth, to obtain the surface rate (面率).
4. Divide the surface rate by half the depth to obtain the water surface diameter.
5. Continue with further multiplications to isolate the rain volume, then divide by the mouth-area (squared and tripled) to obtain the flat-ground rain depth.

Mathematical summary: The heavenly pool basin is a frustum of a cone. The rain volume collected is:

$$V_{\text{rain}} = \frac{\pi h_{\text{rain}}}{3} \left(R^2 + Rr + r^2\right)$$

where R is the top radius, r is the base radius, and h_{rain} is the rain depth. Dividing by the mouth area πR^2 gives the equivalent flat-ground rain depth.

2.2 Problem 7: Measuring Rain with a Round Vessel

圓罌測雨 (Yuan Ying Ce Yu)

問 Using a round vessel (圓罌) to catch rain:

- Mouth diameter (口徑): 1 *chi* 5 *fen* (10.5 *cun*)
- Belly diameter (腹徑): 2 *chi* 4 *cun* (24 *cun*)
- Base diameter (底徑): 8 *cun*
- Depth (深): 1 *chi* 6 *cun* (16 *cun*)
- Rainwater depth inside: 1 *chi* 2 *cun* (12 *cun*)

Using the "dense rate" (密率, $\pi \approx 22/7$) for circle computations, what is the flat-ground rain depth?

答曰 Flat-ground rain depth: **1 *chi* 8 *cun* plus $\frac{6483}{74088}$ *cun*.** (Editor's corrected value.)

術曰曰 Multiply the base diameter by the belly diameter, and also square each; combine these three values. Multiply by half the vessel depth; multiply by 11 and divide by 42 to obtain the lower volume.

Take half the vessel depth plus the rain depth, subtract from the total depth to obtain the upper depth. Subtract the belly diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth; add the mouth-diameter times half-depth to obtain the surface rate.

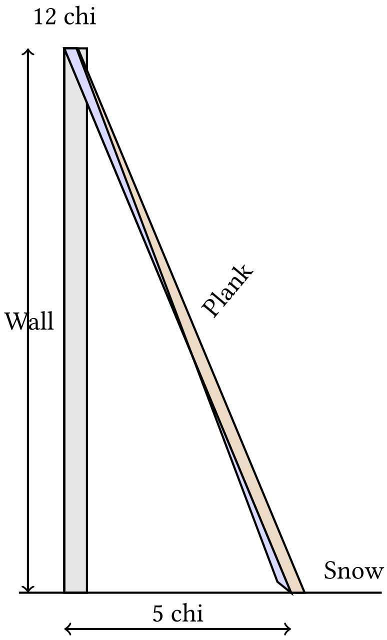
Multiply the surface rate by the belly rate; also square each and combine; multiply by 11 and divide by 42 to obtain the upper rate. Combine the two rates (upper and lower) to obtain the total rain volume as dividend. Divide by three times the squared mouth-diameter to obtain the flat-ground rain depth.

按 (Editor's note:) The original problem statement mentions "dense rate for circle computation" which is irrelevant to finding the flat-ground rain depth. If one were seeking the vessel's rain volume, then this phrase would be appropriate. There were errors in the original computation; the above presents the corrected method.

2.3 Problem 8: Verifying Snow Accumulation

峻積驗雪 (Jun Ji Yan Xue)

問 To verify snow for predicting the harvest: A wall is 1 *zhang* 2 *chi* (12 *chi*) high. A plank leans against it, the base 5 *chi* from the wall's foot, with the tip level with the wall top. Snow accumulates 4 *cun* thick on the plank. The steep accumulation is thin while the flat accumulation is thick. What is the flat-ground snow depth?



答曰 Flat-ground snow depth: **1 *chi* 4 *fen***.

術曰曰 Use the method of Shao Guang (少廣), with the connected-branch (連枝) method:

1. Square the distance from the base (去址): this is the corner (隅).
2. Square the wall height; combine with the corner above.
3. Square the snow thickness; multiply by the combined value above to obtain the dividend.
4. (Reduce if possible.) Extract the connected-branch square root to obtain the flat-ground snow depth.

草曰曰 Convert all measurements to *cun*:

- Distance from base: 50 *cun*; squared = 2,500 (corner).
- Wall height: 120 *cun*; squared = 14,400.
- Combined: 14,400 + 2,500 = 16,900.
- Snow thickness: 4 *cun*; squared = 16.
- Multiply: $16{,}900 \times 16 = 270{,}400$ (dividend).

The corner and dividend share a factor of 100; reduce to get dividend 2,704 and corner 25. Extracting the square root: $\sqrt{2704/25} = 52/5 = 10.4$ *cun* = **1 *chi* 4 *fen***.

按 (Editor's note:) Both the principle and method are correct. In fact, this uses the right-triangle (勾股) method, though not named as such. The "connected-branch" terminology conceals the underlying principle. The diagram illustrates: AB is the flat-ground snow depth (equal to the wall-top snow depth), BC is the excess snow on the plank. Triangle ABC is similar to the triangle formed by the plank leaning against the wall. The wall height is the large leg, the distance from base is the large base, the plank is the large hypotenuse. The flat-ground snow depth is found by proportional reasoning via the Pythagorean theorem.

2.4 Problem 9: Verifying Snow with a Bamboo Container

竹器驗雪 (Zhu Qi Yan Xue)

問 Using a round bamboo container (圓竹器) to verify snow:

- Mouth diameter (口徑): 1 *chi* 6 *cun* (16 *cun*)
- Depth (深): 1 *chi* 7 *cun* (17 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)

Snow falls into it, filling to a depth of 1 *chi*. The container is permeable to wind; when much snow enters, the flat-ground depth is less. What is the flat-ground snow thickness?

答曰 Flat-ground snow thickness: **9 *cun* plus $\frac{764}{3439}$ *cun***.

術曰曰 Subtract the base diameter from the mouth diameter; multiply by the snow depth, halve, and square to obtain the corner (隅). Square the container depth and multiply by the squared snow depth; combine with the corner and multiply by the squared snow depth to obtain the dividend. Extract the connected-branch cube root (三乘方) to obtain the flat-ground snow thickness.

草曰曰 Convert all measurements to *cun*:

- Mouth diameter: 16 *cun*; base diameter: 12 *cun*.
- Difference: 4 *cun*; snow depth: 10 *cun*.
- Multiply: $4 \times 10 = 40$; halve = 20 (this is the upper-*lian* 上廉).
- Container depth squared: $17^2 = 289$.
- Snow depth squared: $10^2 = 100$.
- Multiply: $289 \times 100 = 28{,}900$; combine with the corner and continue the cube-root extraction.

The working proceeds through the Chinese "method of extracting cube roots with carrying" (三乘方開法), yielding the flat-ground snow depth of 9 *cun* with remainder 764/3439.

按 (Editor's note:) The phrase "permeable to wind" does not affect the computation method. Some commentators suggest the snow depth in the container differs from the flat-ground depth due to wind effects; this problem uses the same geometric principle as the Heavenly Pool rain problem, adapted for a frustum-shaped bamboo container.

Summary of Fascicle 2 Problems

No.	Title	Pinyin	Topic
1	推氣治歷	Tui Qi Zhi Li	Computing solar terms from two winter solstice observations
2	治歷推閏	Zhi Li Tui Run	Finding intercalation parameters for a calendar
3	治歷演紀	Zhi Li Yan Ji	Deriving 23 epoch parameters using Dayan method
4	綴術推星	Zhui Shu Tui Xing	Computing Jupiter's motion rates
5	揆日究微	Kui Ri Jiu Wei	Comparing solar shadow lengths at two locations
6	天池測雨	Tian Chi Ce Yu	Measuring rain with a conical basin
7	圓罌測雨	Yuan Ying Ce Yu	Measuring rain with a round vessel
8	峻積驗雪	Jun Ji Yan Xue	Converting snow on a slanted plank to flat-ground depth
9	竹器驗雪	Zhu Qi Yan Xue	Converting snow in a bamboo container to flat-ground depth

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End of Fascicle 2

"Seven planets revolve in the firmament; the affairs of men are tracked by pursuing and connecting [their motions].
Stars by night, gnomon by day—as ages pass, [methods] become coarse;
only the wise can reform them."
— Qin Jiushao, Preface to *Shuxue Jiuzhang*

Siku Quanshu Edition

Shu Xue Jiu Zhang

Fascicles 3–4

Tian Yu (Fields) and Ce Wang (Surveying)

Author: Song Dynasty, Qin Jiushao

Dates: c. 1202–1261

Topics: Juan 3 Shang/Xia: Tian Yu (Fields)
Juan 4 Shang/Xia: Ce Wang (Surveying)

Typeset from the Siku Quanshu (Si Ku Quan Shu) edition
Modern edition with English annotations

Contents

Introduction

The *Shu Xue Jiu Zhang* (Mathematical Treatise in Nine Chapters), written by Qin Jiushao during the Southern Song Dynasty, is one of the greatest masterpieces of medieval Chinese mathematics. The work contains nine chapters, each devoted to a different class of problems.

This volume presents Fascicles 3 and 4 of the work, covering:

- **Juan 3 Shang – Tian Yu (Fields, Part 1)**: Problems on area measurement, including squares, circles, triangles, and composite figures. This section demonstrates Qin's algebraic techniques for solving geometric problems involving fields of various shapes.
- **Juan 3 Xia – Tian Yu (Fields, Part 2)**: Continuation of field problems, including more complex configurations with counting rod diagrams.
- **Juan 4 Shang – Ce Wang (Surveying, Part 1)**: Problems on remote measurement and surveying, using similar triangles and the hook-and-gourd (gou gu) method. Contains numerous counting rod computational tables.
- **Juan 4 Xia – Ce Wang (Surveying, Part 2)**: Further surveying problems, including measurement of heights, distances, and inaccessible objects.

The text follows the classical format of Chinese mathematical literature:

- **Wen**: The problem statement
- **Da Yue**: The answer
- **Shu Yue**: The method/algorithm
- **Cao Yue**: The working/calculation
- **Tu**: Diagrams and illustrations

1 Juan 3 Shang – Tian Yu (Fascicle 3, Part 1: Fields)

Introductory note: This fascicle treats problems on area measurement (Tian Yu). The problems involve squares, circles, triangles, and composite figures. Qin Jiushao employs his "method of the celestial element" (Tian Yuan Shu) alongside traditional algorithmic techniques. The problems are noted for their algebraic complexity and elegant solutions.
From the original text:

> An this juan yi fang yuan xie zhi mi ji xiang qiu, ji fang tian shao guang gou gu zhu fa er shu zhong lei cheng lei chu cuo zong bian huan yu chang fa hui ran qi ben ze chu yu li tian yuan yi fa, jin ze qi nan jie zhe yi li tian yuan yi fa ming zhi, jie bu gong zi po yi.
>
> (*The methods of this fascicle involve squares, circles, oblique and straight lines, areas and volumes, all sought through mutual relations. The fundamental principles derive from the method of the celestial element. Those problems difficult to solve by conventional methods are all resolved effortlessly when illuminated by the celestial element method.*)

Problem 1: Square with Inscribed Circular Pond (Fang Zhong Gu Yuan)

Wen Di Yi

Fang Zhong Gu Yuan Chi

You fang zhong gu yuan chi, yin pi bei yu yi jiao, cong wai fang yu xie zhi nei yuan bian qi chi liu cun. Yu jiu gu ji xiu zhi, yu qiu yuan fang, fang xie ge ji he.

Translation: There is a square field with an ancient circular pond inscribed within it. The northern corner has partially collapsed. From the corner of the outer square to the edge of the inner circle, the diagonal distance is 7 chi 6 cun (7.6 chi). Wishing to repair it according to the original design, find: the diameter of the circle, the side of the square, and the diagonal of the square.

Da Yue:

Chi yuan jing san zhang liu chi liu cun, si bai er shi jiu fen cun zhi si bai yi shi er.

Fang mian san zhang liu chi liu cun, si bai er shi jiu fen cun zhi si bai yi shi er.

Fang xie wu zhang yi chi ba cun, si bai er shi jiu fen cun zhi si bai yi shi er.

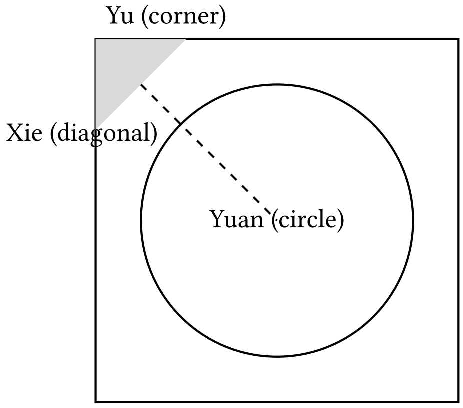
Answer: The pond's diameter is 36 chi 6 cun plus $\frac{412}{429}$ cun; the square's side is 36 chi 6 cun plus $\frac{412}{429}$ cun; the square's diagonal is 51 chi 8 cun plus $\frac{412}{429}$ cun.

Shu Yue:

Yi shao guang qiu zhi, tai tai shu. Ru zhi xie zi cheng bei zhi wei shi, bei xie wei yi fang, yi ban cun wei cong yu. Kai ping fang de jing.

Method: Using the "lesser breadth" (shao guang) method with the "transformation of the embryo" (tai tai) technique. Square the diagonal and double it to obtain the dividend (shi). Take twice the diagonal as the negative coefficient of the linear term (yi fang). Take half a cun as the coefficient of the quadratic term (cong yu). Extract the square root to find the diameter.

Tu (Diagram of the Ancient Pond):



Problem 2: Triangular Field (San Xie Qiu Ji)

Wen Di Er

San Xie Qiu Ji

You san jiao xing tian, da xie yi shi san li, xiao xie yi shi si li, zhong xie yi shi wu li. Li fa san bai bu, yu zhi wei tian ji he.

Translation: There is a triangular field. The large diagonal (da xie) is 13 li, the small diagonal (xiao xie) is 14 li, and the middle diagonal (zhong xie) is 15 li. With 300 bu per li, find the area.

Da Yue:

Tian ji san bai yi shi wu qing.

Answer: The area is 315 qing.

Shu Yue:

Yi shao guang qiu zhi. Yi xiao xie mi bing da xie mi, jian zhong xie mi, yu ban zhi, zi cheng yu shang. Yi xiao xie mi cheng da xie mi, jian shang, yu si yue zhi wei shi. Yi wei cong yu, kai ping fang de ji.

Method: Using the "lesser breadth" method. Add the squares of the small and large diagonals, subtract the square of the middle diagonal. Halve the remainder and square it (placed above). Multiply the square of the small diagonal by the square of the large diagonal; subtract the result above; divide the remainder by 4 to get the dividend. Take 1 as the coefficient of the quadratic term; extract the square root to find the area.

In modern notation: If $a = 13$, $b = 14$, $c = 15$, then the area is:

$$S = \sqrt{\frac{1}{4}\left[a^2b^2 - \left(\frac{a^2+b^2-c^2}{2}\right)^2\right]}$$

This is equivalent to **Heron's formula**:

$$S = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

Problem 3: Square Field with Central Circular Pond (Fang Tian Yuan Chi)

Wen Di San

Fang Tian Yuan Chi

You fang tian yi duan, nei you yuan chi shui zhan zhi, wai ji di san qian yi bai liu shi ba bu. Zhi yun cong nei chi zhou zhi wai tian jiao xie er shi yi bu ban, nei wai fang yuan ge ji he.

Translation: There is a square field with a circular pond inside it. The remaining land outside the pond amounts to 3168 bu. It is given that the diagonal distance from the edge of the inner pond to the corner of the outer square is 21.5 bu. Find the dimensions of both the inner circle and outer square.

Da Yue:

Nei chi zhou si shi er bu, jing yi shi san bu ban, fang tian mian liu shi bu.

Answer: The inner pond's circumference is 42 bu, its diameter is 13.5 bu; the square field's side is 60 bu.

Shu Yue:

Yi shao guang qiu zhi. Zhi ji yi yuan lu san yue zhi wei shi, bei xie wei cong fang, xie mi wei yi yu, kai ping fang de nei jing. Jia bei xie de fang mian.

Method: Using the "lesser breadth" method. Place the area and divide by the circular rate 3 to obtain the dividend. Twice the diagonal gives the linear coefficient. The square of the diagonal gives the negative quadratic coefficient. Extract the square root to find the inner diameter. Add twice the diagonal to obtain the square's side.

Problem 4: The Drowned Field (Piao Tian)

Wen Di Si

Piao Tian

You piao tian yi duan, zhong xie yi shi liu bu, shui zhi wu bu, jian zhong xie yi shi liu yu yi shi yi bu wei yuan da xie, nei jian can da xie er shi bu, yu jiu bu yi shi yi bu zhi yi wei yuan da xie.

Translation: There is a "drowned field" (a triangular field partially submerged). The middle diagonal is 16 bu; the water depth is 5 bu. Subtracting 5 from 16 gives 11 bu as the base large diagonal.

Da Yue:

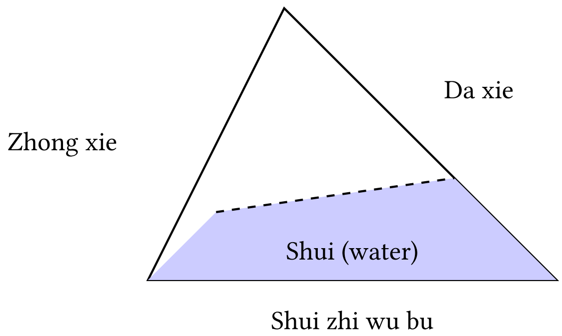
Tian ji ruo gan.

Shu Yue:

Cao yue: yi shui zhi wu jian zhong xie yi shi liu yu yi shi yi bu wei fa, yi zhong xie yi shi liu cheng da can er shi de san bai er shi wei da xie shi, yi fa chu zhi de er shi jiu bu yi shi yi fen bu zhi yi wei yuan da xie. Nei jian can da xie er shi bu, yu jiu bu yi shi yi bu zhi yi. You yi xia xiao xie yi shi yi bu wei shui da xie, yi fa yi shi yi cheng jing yi bu yi shi yi fen bu zhi yi wei shui xiao xie, yi fa yi shi yi cheng shui zhi wu bu de wu shi wu bu, yi shui da xie yi shi yi bu cheng shui xiao xie yi shi er bu de yi bai san shi er bu wei shi, yi fa yi shi yi zi cheng de yi bai er shi yi bu wei fa, chu shi de yi bu yi shi yi fen bu zhi yi wei shui ji.

Method: (Detailed calculation with specific numerical operations as recorded in the original text.)

Tu (Diagram of the Drowned Field):


Da xie
Zhong xie
Shui (water)
Shui zhi wu bu

2 Juan 3 Xia – Tian Yu Xu (Fascicle 3, Part 2: Fields Continued)

Problem 5: Division of Land Among Three Families (San Hu Fen Tian)

Wen Di Wu

San Hu Fen Tian

You tian yi duan, yu fen yu jia yi bing san hu. Jia duo yi bai wu shi bu, yi duo yi bai er shi bu, bing duo jiu shi bu. San hu gong tian ruo gan, ge de ji he.

Translation: There is a field to be divided among three families jia, yi, and bing. Family jia receives 150 bu more than the base amount; family yi receives 120 bu more; family bing receives 90 bu more. If the total area is given, find each family's share.

Da Yue:

Jia tian ruo gan, yi tian ruo gan, bing tian ruo gan.

Shu Yue:

Yi shao guang qiu zhi. Bing san hu duo chu zhi shu yi jian zong ji, yu yi san hu yue zhi, ge de ben ji. You ge jia duo chu zhi shu ji de ge hu zhi tian.

Method: Using the "lesser breadth" method. Add the excess amounts from all three families and subtract from the total area. Divide the remainder by 3 to get the base amount for each. Add each family's excess to the base amount to obtain each share.

Problem 6: Circular and Square Fields (Yuan Tian Fang Tian)

Wen Di Liu

Yuan Tian Fang Tian

You fang yuan tian ge yi duan, gong ji yi qian er bai ba shi bu. Zhi yun fang mian ru yuan jing shao si bu, wen fang yuan mian jing ge ji he.

Translation: There is a square field and a circular field. Their total area is 1280 bu. It is given that the square's side is 4 bu less than the circle's diameter. Find the side and diameter.

Da Yue:

Fang mian ruo gan, yuan jing ruo gan.

Shu Yue:

Yi shao guang qiu zhi. Zhi gong ji yi yuan lu san, fang lu si ge yue zhi wei shi, fang mian shao yuan jing si bu wei cong fang, kai ping fang de yuan jing. Jian si de fang mian.

Method: Using the "lesser breadth" method. Place the combined area; divide by the circular rate 3 and square rate 4 to obtain the dividend. The difference of 4 bu between square side and circle diameter gives the linear coefficient. Extract the square root to find the diameter. Subtract 4 to obtain the square's side.

Problem 7: Circular Segment Field (Hu Tian)

Wen Di Qi

Hu Tian

You hu tian yi duan, xian wu shi bu, shi er shi wu bu, qiu ji ruo gan.

Translation: There is a circular segment field (hu tian) with chord (xian) 50 bu and arrow (shi) 25 bu. Find its area.

Da Yue:

Tian ji ruo gan.

Shu Yue:

Yi xian shi qiu zhi. Yi xian cheng shi, you yi shi zi cheng, bing zhi, ban zhi wei shi. Kai ping fang de ji.

Method: Using the chord-arrow method. Multiply the chord by the arrow; add the square of the arrow; halve to obtain the dividend. Extract the square root to find the area.

Note: For a circular segment with chord c and sagitta s, Qin's formula gives:

$$A = \frac{cs + s^2}{2}$$

This is an approximation to the true area of a circular segment.

Counting Rod Diagram for the Calculation:

Xian	Shi	Xian Cheng Shi	Shi Mi
50	25	1250	625
He		**Ban Zhi**	**Ji**
1875		937.5	≈ 30.6

3 Juan 4 Shang – Ce Wang (Fascicle 4, Part 1: Surveying)

Introductory note: This fascicle treats problems on surveying and remote measurement (Ce Wang). These problems involve measuring inaccessible distances and heights using the "hook-and-gourd" (gou gu) method and similar triangles. The techniques demonstrated here are foundational to ancient Chinese geodesy and cartography.

Problem 1: The Round City (Yuan Cheng)

Wen Di Yi

Yuan Cheng

You yuan cheng bu zhi zhou jing, si men zhong kai, bei wai san li you qiao mu chu nan men bian zhe dong men jiu li nai jian mu. Yu zhi cheng zhou jing ge ji he.

Translation: There is a circular city of unknown circumference and diameter. It has four gates at the cardinal points. Three li north of the north gate stands a tall tree. Going out the south gate and then turning east for 9 li, the tree becomes visible. Find the circumference and diameter of the city.

Da Yue:

Jing jiu li, zhou er shi qi li.

Answer: The diameter is 9 li, the circumference is 27 li.

Shu Yue:

Yi gou gu xi jie qiu zhi. Yi wei cong yu, wu yin bei li wei cong qi lian, zhi bei li mi ba yin wei cong wu lian, yi bei li cheng shang lian wei yi cong, san lian bei fu lu cheng wu lian, wei yi shang lian, yi bei li cheng shang lian wei shi, kai ling long jiu cheng fang de shu, zi cheng wei jing, yi san yin jing de zhou.

Method: Using the "hook-and-gourd evening-peak" method. Take 1 as the leading coefficient. Multiply the north distance by 5 for the seventh-degree coefficient. Place the square of the north distance, multiply by 8 for the fifth-degree coefficient. Multiply the north distance by the upper coefficient for the negative linear coefficient. Double the negative rate to form the fifth-degree coefficient as the negative upper coefficient. Multiply the north distance by the upper coefficient for the dividend. Extract the ninth-degree root (ling long kai fang) to obtain the number; square it for the diameter. Multiply the diameter by 3 for the circumference.

Problem 2: Measuring a Tower (Ce Ta)

Wen Di Er

Ce Ta

You ta gao wei zhi qi shu, qu ta ruo gan bu li yi biao, ren mu shang wang jian ta jian, tui xing ruo gan bu you li yi biao, ren mu shang wang yi jian ta jian. Qiu ta gao ji qu ta yuan jin.

Translation: A tower of unknown height stands at an unknown distance. At a certain distance from the tower, a pole is erected; looking upward from eye level, the tower's peak is visible. Retreating a certain distance and erecting another pole, the peak is again visible from eye level. Find the tower's height and distance.

Da Yue:
Ta gao ruo gan, qu ta yuan ruo gan.

Shu Yue:
Yi gou gu qiu zhi. Liang biao tui xing zhi cha wei fa, qian biao qu ta yuan cheng biao gao, hou biao qu ta yuan cheng biao gao, xiang jian wei shi, shi ru fa er yi de ta gao. You yi fa chu qian biao qu ta yuan cheng liang biao gao cha jia biao gao de ta gao.

Method: Using the hook-and-gourd method. The difference between the retreat distances of the two poles is the divisor. Multiply the front pole's distance to the tower by the pole height; multiply the rear pole's distance by the pole height; subtract to obtain the dividend. Divide to get the tower height. Alternatively, multiply the front distance by the difference of pole heights, divide by the divisor, and add the pole height.

In modern notation: Let d_1 = front distance, d_2 = rear distance, h = pole height. Then:

$$\text{Tower height} = \frac{d_2 \cdot h - d_1 \cdot h}{d_2 - d_1} = h$$

This uses the principle of similar triangles, where the ratio of heights equals the ratio of distances.

Problem 3: The Square City (Fang Cheng)

Wen Di San

Fang Cheng

You fang cheng yi zuo, bu zhi da xiao. Bei men wai you shu, chu nan men zhi xing ruo gan bu zhe er xi xing ruo gan bu jian shu. Qiu cheng fang mian.

Translation: There is a square city of unknown size. Outside the north gate stands a tree. Going out the south gate and walking straight a certain number of steps, then turning west and walking a certain number of steps, the tree becomes visible. Find the side of the square city.

Da Yue:
Cheng fang mian ruo gan bu.
Answer: The city's side is a certain number of bu.

Shu Yue:
Yi gou gu qiu zhi. Liang xing bu xiang cheng wei shi, bing liang xing bu wei cong fang, kai ping fang de cheng fang mian.

Method: Using the hook-and-gourd method. Multiply the two walk distances for the dividend. Add the two walk distances for the linear coefficient. Extract the square root to obtain the city's side.

If a = southward steps and b = westward steps:

$$x^2 + (a+b)x = ab \quad \Rightarrow \quad x = \frac{-(a+b) + \sqrt{(a+b)^2 + 4ab}}{2}$$

Problem 4: Counting Rod Root Extraction (Chou Suan Kai Fang)

Wen Di Si

Chou Suan Kai Fang

Wen yi kai fang shu jie gao ci fang cheng, qi chou bu lie zhi fa ruo he.

Translation: On the method of extracting roots to solve higher-degree equations, how are the counting rods arranged?

Shu Yue:

Kai fang zhi fa, xian lie shi yu zhong, ci lie yu yu xia, ci lie fang yu shang. Shang chu zhi, yi shang cheng yu sheng lian, yi shang cheng lian sheng fang, yi shang cheng fang chu shi. Shi jin ze zhi, bu jin ze bei fang, san lian wei fang fa, xu shang chu zhi.

Method: In the method of root extraction, first place the dividend (shi) in the middle, then place the unit coefficient (yu) below, and the linear coefficient (fang) above. Divide by estimation; multiply the estimate by the unit coefficient to generate the linear coefficient; multiply the estimate by the linear coefficient and subtract from the dividend. When the dividend is exhausted, stop. If not exhausted, double the linear coefficient and proceed to the next digit.

Counting Rod Layout for Root Extraction:

Shang	Shi (Dividend)	Fang (Linear)	Lian	Yu (Unit)
7	360	130	...	1
Iterative extraction process				

Note: In the counting rod system, vertical strokes represent units $(1, 2, 3, 4)$ and horizontal strokes represent fives. The symbol T represents 6 (a horizontal stroke over a vertical one). The diagram shows the iterative process of root extraction.

4 Juan 4 Xia – Ce Wang Xu (Fascicle 4, Part 2: Surveying Continued)

Problem 5: Measuring Depth of Water (Ce Shui Shen)

Wen Di Wu

Ce Shui Shen

You jing bu zhi qi shen, yi sheng ce zhi, sheng chang bei yu jing shen, yu san chi; san zhe er ce zhi shi zu. Qiu jing shen ji sheng chang.

Translation: There is a well of unknown depth. Measuring with a rope, the rope's length is twice the well's depth with 3 chi to spare. Folding the rope in three and measuring, it fits exactly. Find the depth of the well and the length of the rope.

Da Yue:

Jing shen san chi, sheng chang jiu chi.

Answer: The well is 3 chi deep; the rope is 9 chi long.

Shu Yue:

Yi ying bu zu qiu zhi. Bei sheng yu san chi wei ying, san zhe shi zu wei shi zu, yi ying cheng shi zu wei shi, bing ying shi zu wei fa, chu zhi de jing shen. Bei zhi jia yu de sheng chang.

Method: Using the "surplus and deficit" (ying bu zu) method. The 3 chi excess when doubled is the surplus; the exact fit when tripled is the exact amount. Multiply the surplus by the exact amount for the dividend; add surplus and exact amount for the divisor; divide to obtain the well depth. Double and add the excess for the rope length.

In modern notation: Let d = well depth, L = rope length.

$$L = 2d + 3, \qquad \frac{L}{3} = d$$

Substituting: $2d + 3 = 3d \Rightarrow d = 3$, $L = 9$.

Problem 6: Measuring the Height of a Pagoda (Ce Ta Gao)

Wen Di Liu

Ce Ta Gao

You ta gao wei zhi, qu ta bai bu li yi biao gao wu chi, ren mu qu biao yi chi ba cun, shang wang ta jian yu biao duan can he. Tui xing liu shi bu, ren mu qu biao duan reng can he ta jian. Qiu ta gao ji ta xin qu biao yuan jin.

Translation: A pagoda of unknown height stands at an unknown distance. 100 bu from the pagoda, a 5-chi pole is erected. The observer's eye is 1.8 chi from the ground; looking up, the pagoda's peak aligns with the pole's tip. Retreating 60 bu, the observer's eye again aligns with both the pole's tip and the pagoda's peak. Find the pagoda's height and its distance from the pole.

Da Yue:

Ta shen gao san zhang, ta gao yi shi yi zhang qi chi, yu ba zhang qi chi wei ta xin mu chang.

Answer: The pagoda's body is 3 zhang high; the total height (from ground to peak) is 11 zhang 7 chi; the distance from the pagoda's center to the measuring point is 8 zhang 7 chi.

Shu Yue:

Ci jie da xiao xing tong shi xiang qiu fa ye. Ren mu qu ta wei zong gu, jiao ren mu qu gan wei fen xiao gu, ren mu shang ta jian gao ta shen jie wei zong gu gao. Xiang lun gao wei fen da gou, shu yi gan qu ta fen da gou yu xiao jiao gan qu ta wei fen da gou.

Method: These are all applications of similar triangles (da xiao xing tong shi). The distance from eye to pagoda is the total height line; the distance from eye to pole is the small height line. The height from eye level to pagoda peak is the total height; the height from eye level to pole top is the small height. The method uses proportional relationships between these quantities.

This problem uses the principle of similar triangles:

$$\frac{\text{Pagoda height} - \text{eye height}}{\text{Distance to pagoda}} = \frac{\text{Pole height} - \text{eye height}}{\text{Distance to pole}}$$

Problem 7: Measuring Distance Across a River (Ge He Ce Yuan)

Wen Di Qi

Ge He Ce Yuan

Wen ge he wang yi gao an, bu zhi qi yuan. Ping di li yi biao, ren mu shang wang jian an ding, you tui li yi biao wang zhi yi jian an ding. Qiu he kuo ji an gao.

Translation: Across a river, a high bank is visible at an unknown distance. A pole is erected on level ground; looking up from eye level, the bank's top is visible. Retreating and erecting another pole, the bank's top is again visible. Find the width of the river and the height of the bank.

Da Yue:

He kuo ruo gan bu, an gao ruo gan chi.

Shu Yue:

Yi chong cha qiu zhi. Liang biao tui xing bu shu zhi cha wei fa, qian biao qu an yuan cheng biao gao wei shi, shi ru fa er yi de an gao. You yi fa chu hou biao qu an yuan cheng biao gao yi de an gao, liang zhe xiang yan.

Method: Using the "double difference" (chong cha) method. The difference between the two retreat distances is the divisor. Multiply the front pole's distance to the bank by the pole height for the dividend; divide to obtain the bank height. Similarly for the rear pole; the two results should agree (providing verification).

Problem 8: Higher-Degree Equations (Jiu Cheng Fang)

Wen Di Ba

Jiu Cheng Fang

Wen yi kai jiu cheng fang shu jie fang cheng, qi li fa zhi yuan ruo he.

Translation: On the method of extracting ninth-degree roots to solve equations — what is the fundamental principle of this method?

Shu Yue:

Kai fang zhi shu, zhi yu jiu cheng fang, qi li yi ye. Xian lie shi yu zhong, ci ding yu wei, ci li fang fa, lian fa. Shang chu zhi ji, yi shang cheng yu sheng lian, yi shang cheng lian sheng fang, yi shang cheng fang chu shi. Mei jin yi wei, ze fang tui yi, lian tui er, yu tui san. Ru shi die jin, zhi zhi shi jin wei zhi.

Method: The method of root extraction, up to the ninth degree, follows the same principle. First place the dividend in the middle; then determine the unit position; then establish the linear and intermediate coefficients. During division-by-estimation: multiply the estimate by the unit coefficient to generate the intermediate coefficients; multiply the estimate by the intermediate coefficients to generate the linear coefficient; multiply the estimate by the linear coefficient and subtract from the dividend. At each decimal shift, the linear coefficient descends one place, the intermediate descend two places, and the unit descends three. Continue iteratively until the dividend is exhausted.

Note: This describes **Horner's method** for solving polynomial equations, which Qin Jiushao developed independently centuries before Horner. For a polynomial:

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

The method systematically reduces the polynomial by factoring out $(x - r)$ where r is the estimated root at each step.

Layout for Ninth-Degree Root Extraction:

Shang	Shi	Fang	1-Lian	2-Lian	3-Lian	4-Lian	5-Lian	6-Lian	Yu
7	–	153	...	...	...	...	...	...	1
Iterative computation with place-value shifts									

Appendix: Mathematical Notes

A.1 The Counting Rod System (Chou Suan)

The counting rod system (Chou Suan) was the primary computational tool in ancient Chinese mathematics. Numbers were represented by bamboo sticks arranged in a decimal place-value system:

- **Units (1–4)**: Vertical strokes |, ||, |||, ||||
- **Fives**: A horizontal stroke $=$ placed over the units
- **6–9**: Combinations, e.g., T (5+1), $\equiv$ (3 horizontal over vertical)
- **Zero**: Represented by a blank space or circle $\bigcirc$

The rods were arranged on a counting board in columns, with each column representing a decimal place. This positional system enabled sophisticated algebraic computations including root extraction and solution of polynomial equations.

A.2 The "Celestial Element" Method (Tian Yuan Shu)

Qin Jiushao was a master of the "celestial element" method, an early form of algebraic symbolism. The unknown quantity was called "celestial element" (Tian Yuan, tian yuan), and equations were set up using counting rod arrangements. This method could solve polynomial equations of degree up to 10, anticipating Horner's method by over 500 years.

A.3 Qin Jiushao's Formula for Triangular Area

For a triangle with sides a, b, c, Qin gave the formula:

$$S = \sqrt{\frac{1}{4}\left[c^2a^2 - \left(\frac{c^2 + a^2 - b^2}{2}\right)^2\right]}$$

This is equivalent to Heron's formula and demonstrates remarkable algebraic insight for the 13th century.

A.4 The "Surplus and Deficit" Method (Ying Bu Zu)

The method of surplus and deficit (Ying Bu Zu Shu) is a general technique for solving linear problems. Given two trial values yielding surplus and deficit results, the true value is found by:

$$x = \frac{a_2b_1 + a_1b_2}{a_1 + a_2}$$

where a_1, a_2 are the trial inputs and b_1, b_2 are the corresponding surplus/deficit amounts. This method can be applied to a wide variety of practical problems.

End of Fascicles 3–4

Shu Xue Jiu Zhang – Juan 3 Shang, Juan 3 Xia, Juan 4 Shang, Juan 4 Xia

數學九章

卷五：**賦役**卷六：**錢穀**

宋秦九韶撰

–

四庫全書

問答術草

卷五上賦役

按：賦役一章，即古九章中差分粟米諸法。但從來算書設題之數未有如此卷之多者。如首題答數至一百七十五條，每條步算之數至十餘位，而得數皆無不合，亦可謂能廣其用矣。且諸問皆足以明貴賤廩税及交質變易之同異，使公私各得其平，誠治民之要務也。但題語多晦，術草中間有與所問不相應者，亦於各條下指出，俾觀者無惑焉。

復邑修賦

問：有海埍縣地，今有復漲，歲从鄉井再成，申諸刱邑。稱土排到六鄉，以附郭為甲，最遠為已，各有田九等，開具下項：

甲鄉共計田十四萬一百九十三畝三角一十二步
上等：上田五千六百七十八畝一角四十八步；中田四千八百九十二畝三十步；下田六千六百二十一畝五十四步 中等：上田八千二百二十五畝二十四步；中田一萬三十五畝六步；下田一萬六千五百三十畝 下等：上田二萬一千九十畝二十四步；中田三萬二千六十畝三步；下田三萬五千六十一畝三角三步

乙鄉田共計八萬四千一十畝二角二步
上等：上田六千七百八十九畝一角三十六步；中田五千九百八十七畝二角；下田八千一十畝三角三步 中等：上田七千五百四十一畝；中田九千一百二十一畝二角一十二步；下田一萬九千六十步 下等：上田一萬八千三十七畝一角六步；中田九千四百五十六畝三角五十九步；下田無

丙鄉田共計一十二萬九百三十五畝五十八步五分
上等：上田四千八百六十八畝二角三步；中田五千九百七十九畝三角六步；下田六千八百八十八畝二角六步 中等：上田七千九百八十四畝一角；中田一萬四千五十六畝一十二步；下田二萬三千三百三十三畝一十二步 下等：上田二萬七千七百五十五畝一十六步五分；中田無；下田三萬七十畝三步

丁鄉田共計八萬九千六十六畝二步三分
上等：上田一萬一千一百二畝一步；中田九千八百七十六畝一角；下田八千七百六十五畝一角三十步 中等：上田七千五百三十九畝三十四步三分；中田無；下田一萬二千九百八十七畝四十二步 下等：上田無；中田五千四百三十二畝一角六步；下田三萬三千三百六十三畝三角九步

戊鄉田共計二十萬四千四百七十四畝一角二十四步四分
上等：上田二萬四千六百三十二畝三十九步；中田一萬三千五百二十一畝二十七步；下田九千九百八十八畝三角三步

戊鄉田共計二十萬四千四百七十四畝一角二十四步四分
中等：上田八千八百七十七畝五十六步四分；中田一萬一千三百三十三畝三角；下田二萬七千六十七畝 下等：上田一萬九千八百七十六畝三角六步；中等田七萬九千一百三十五畝三角四十三步；下田一萬四十一畝二角三十步

已鄉田共計一十五萬八千四百六十畝三角一十八步二分
上等：上田無；中田七千七百八十八畝三角五十一步；下田無 中等：上田九千九百九十九畝二角三步；中田一萬八百三十六畝五十六步；下田無 下等：上田三萬二千八十九畝一角四十五步六分；中田四萬三千六百七十八畝二角五十七步；下田五萬四千六十七畝三角四十五步六分

照得昨來本縣原科苗米一十萬三千五百六十七石八斗四升四合三勺，和買一萬三千四百九十八匹一丈七尺三寸七分六釐，夏稅九千八百七十六匹三丈二尺六寸五分八釐。其六鄉田係三色，甲為上，乙丙為次，丁戊己又為次。先令官物為三差，使上比中、中比下皆十分外差一，次令各鄉九等皆於十分內差一，拋科用合租額。其乙鄉田最肥，次丁，次甲，次丙，次己，次戊。欲知三色九等每畝等則及共科數各鄉幾何?
［按］題內言田有三色，甲為上云云，又言乙鄉田最肥云云，語若相左。蓋前言三色者六鄉共科之等，後言田肥者每畝所出之異也。若易田係三色為科有三差，則語意明矣。
答曰：甲鄉：上等上田苗三斗二升三合二勺，和一尺六寸九分，稅一尺二寸三分；中田苗二斗九升九勺，和一尺五寸二分，稅一尺一寸一分；下田苗二斗五升八合六勺，和一尺三寸五分，稅九寸九分。
中等上田苗二斗二升六合二勺，和一尺一寸八分，稅八寸六分；中田苗一斗九升三合九勺，和一尺一分，稅七寸四分；下田苗一斗六升一合六勺，和八寸四分，稅六寸二分。
下等上田苗一斗二升九合三勺，和六寸七分，稅四寸九分；中田苗九升六合九勺，和五寸一分，稅三寸七分；下田苗六升四合六勺，和三寸四分，稅二寸五分。
乙鄉：上等上田苗三斗六升三合三勺，和一尺八寸九分，稅一尺三寸九分；中田苗三斗二升六合九勺，和一尺七寸，稅一尺二寸五分；下田苗二斗九升六勺，和一尺五寸二分，稅一尺一寸一分。
中等上田苗二斗五升四合三勺，和一尺三寸三分，稅九寸七分；中田苗二斗一升八合，和一尺一寸四分，稅八寸三分；下田苗一斗八升一合六勺，和九寸五分，稅六寸九分。
下等上田苗一斗四升五合三勺，和七寸六分，稅五寸五分；中田苗一斗九合，和五寸七分，稅四寸二分；下田苗七升二合七勺，和三寸八分，稅二寸八分。
丙鄉：上等上田苗三斗三合五勺，和一尺五寸八分，稅一尺一寸六分；中田苗二斗七升三合一勺，和一尺四寸二分，稅一尺四分；下田苗二斗四升二合八勺，和一尺二寸七分，稅九寸三分。
中等上田苗二斗一升二合四勺，和一尺一寸一分，稅八寸一分；中田苗一斗八升二合一勺，和九寸五分，稅六寸九分；下田苗一斗五升一合七勺，和七寸九分，稅五寸八分。
下等上田苗一斗二升一合四勺，和六寸三分，稅四寸六分；中田苗九升一合，和四寸七分，稅三寸五分；下田苗六升七勺，和三寸二分，稅二寸三分。
丁鄉：上等上田苗三斗四升三合二勺，和一尺七寸九分，稅一尺三寸一分；中田苗三斗八合九勺，和一尺六寸一分，稅一尺一寸八分；下田苗二斗七升四合六勺，和一尺四寸三分，稅一尺五分。
中等上田苗二斗四升二勺，和一尺二寸五分，稅九寸二分；中田苗二斗五合九勺，和一尺七分，稅七寸九分；下田苗一斗七升一合六勺，和八寸九分，稅六寸五分。
下等上田苗一斗三升七合三勺，和七寸二分，稅五寸二分；中田苗一斗三合，和五寸四分，稅三寸九分；下田苗六升八合六勺，和三寸六分，稅二寸六分。
戊鄉：上等上田苗一斗五升三合八勺，和八寸，稅五寸九分；中田苗一斗三升八合四勺，和七寸二分，稅五寸三分；下田苗一斗二升三合一勺，和六寸四分，稅四寸七分。

中等上田苗一斗七合七勺，和五寸六分，稅四寸一分；中田苗九升二合三勺，和四寸八分，稅三寸五分；下田苗七升六合九勺，和四寸，稅二寸九分。
下等上田苗六升一合五勺，和三寸二分，稅二寸三分；中田苗四升六合一勺，和二寸四分，稅一寸八分；下田苗三升八勺，和一寸六分，稅一寸二分。
已鄉：上等上田苗二斗八升二合二勺，和一尺四寸七分，稅一尺八分；中田苗二斗五升三合九勺，和一尺三寸二分，稅九寸七分；下田苗二斗二升五合七勺，和一尺一寸八分，稅八寸六分。
中等上田苗一斗九升七合五勺，和一尺三分，稅七寸五分；中田苗一斗六升九合三勺，和八寸八分，稅六寸五分；下田苗一斗四升一合一勺，和七寸四分，稅五寸四分。
下等上田苗一斗一升二合九勺，和五寸九分，稅四寸三分；中田苗八升四合六勺，和四寸四分，稅三寸二分；下田苗五升六合四勺，和二寸九分，稅二寸二分。
各鄉共科數：甲鄉苗米一萬九千五百五十石二斗四升八合三勺；和買一萬一百九十二丈二尺六寸五分六釐；夏稅七千四百五十七丈六尺八寸九分八釐。
乙鄉苗米一萬七千七百七十二石九斗五升三合；和買九千二百六十五丈六尺九寸六分；夏稅六千七百七十九丈七尺一寸八分。
丙鄉苗米一萬七千七百七十二石九斗五升三合；和買九千二百六十五丈六尺九寸六分；夏稅六千七百七十九丈七尺一寸八分。
丁鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。
戊鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。
已鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。
術曰：以衰分求之。先列本縣色位，自下錐行列之，又以鄉數對列而乘之，副併為法，以除官物數，得一分之率。以率數乘未併者，各得諸鄉之數。次列各鄉等位，自上等置十分，每以內分錐行九折之，至九等止，又各以畝步乘之，副併為鄉法，以除諸各鄉所得官物數，所得為一分之率，以乘未併者，各得每畝稅色。
草曰：列本縣色位三，自下色例十分中十一分，上比中身外加一，上得一百二十一，中得一百一十，下得一百，為上中下三率，列之右行。乃列甲一對上率，乙丙共二對中率，丁戊己共三對下率，列左行。乃以左行列數各相對乘右行率數，其上得一百二十一，其十得二百二十，其下得三百，乃副置而併之，得六百四十一為法。置元科苗米一萬三千五百六十七石八斗四升四合三勺為寔，以法除之，得一百六十一石五斗七升二合三勺為一分之率。以未併下率一百乘率，得一萬六千一百五十七石二斗三升為丁戊己三鄉各科數。以於身下加一，得一萬七千七百七十二石九斗五升三合為乙丙二鄉各科數。又於身下加一，得一萬九千五百五十石二斗四升八合三勺為甲合科數。求和買亦用六百四十一為法，置元科和買一萬三千四百九十八匹，以匹法四丈通之，得五萬三千九百九十二丈，內零一丈七尺三寸七分六釐，得五萬三千九百九十三丈七尺三寸七分六釐為寔，以法除之，得八十四丈二尺三寸三分六釐為一分之率。亦以未併下率一百乘之，得八千四百二十三丈三尺六寸為丁戊己三鄉各科數。次於身下加一，得九千二百六十五丈六尺九寸六分為乙丙二鄉各科數。又於身下加一，得一萬一百九十二丈二尺六寸五分六釐為甲鄉合科數。求夏稅亦置六百四十一為法，置元科九千八百七十六匹，以匹法四丈通之，得三萬九千五百四丈，內零三丈二尺六寸五分八釐，得三萬九千五百七丈二尺六寸五分八釐為寔，以法除之，得六十一丈六尺三寸三分八釐為一分率。亦以未併下率一百乘之，得六千一百六十三丈三尺八寸為丁戊己三鄉各科數。次於身下加一，得六千七百七十九丈七尺一寸八分為乙丙二鄉各科數。又於身下加一，得七千四百五十七丈六尺八寸九分八釐為甲鄉數。

圍田租畝

問：有興復圍田已成，共計三千二十一頃五十一畝一十五步，分三等。其上等每畝起租六斗，中等四斗五升，下等四斗。中田多上田弱半，不及下田太半。欲知三色田畝及各租幾何?
［按］題言中田多上田弱半，不及下田太半。蓋以弱半為三分之一，太半為三分之二也。多上田弱半者，即比上田多三分之一也。不及下田太半者，即比下田少三分之二也。是上田加三分之一為中田，即下田三分之一也。轉言之，則中田為下田三分之一，上田為中田四分之三也。
答曰：上田四百七十七頃八畝一十五步，米二萬八千六百二十四石八斗三升七合五勺；中田六百三十六頃一十畝三角，米二萬八千六百二十四石八斗三升七合五勺；下田一千九百八頃三十二畝一角，米七萬六千三百三十二石九斗。
術曰：以衰分求之。列母子求田率，副併為法，以共田為寔，寔如法而一，得一分之率。以徧乘未併者，得三等田。各以起租乘之，各得米。
草曰：置弱半母四為中率，子三為上率。以太半子二減母三，餘一，以乘中率四，只得四為中泛。又以餘一乘上率三，只得三為上泛。次以太半母三乘中泛四，得一十二為下泛。副併三泛，得一十九為法。置田三千二十一頃五十一畝，以畝法二百四十通之，得七千二百五十一萬六千二百四十步，內子一十五步，得七千二百五十一萬六千二百五十五步為寔。以法一十九除之，得三百八十一萬六千六百四十五步為一分之數。以上泛三因之，得一千一百四十四萬九千九百三十五步為上積。又以中泛四因之一分數，得一千五百二十六萬六千五百八十步為中積。又以下泛一十二乘一分數，得四千五百七十九萬九千七百四十步為下積。其三積各以畝法二百四十約之為畝。其上田得四百七十七頃八畝一十五步，其中田得六百三十六頃一十畝三角，其下積得一千九百八頃三十二畝一角。各以起租，三積為三寔。其上積一千一百四十四萬九千九百三十五步乘上積六斗，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺為上田租。其中積一千五百二十六萬六千五百八十步乘中田租四斗五升，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺。其下積四千五百七十九萬九千七百四十步乘下租四斗，得一千八百三十一萬九千八百九十六石為寔，以畝法二百四十除之，得七萬六千三百三十二石九斗為下田米。
［按］草內求各衰數，意謂以三分為上田衰數，加弱半三分之一得四分為中田衰數，三因中田衰數得一十二分為下田衰數，併之得一十九分為總衰數。其法本屬顯明，但語中參入母子副泛等名目，其意反晦矣。

卷五下賦役

戶稅移割

問：某縣據甲稱：本户田地原納苗三十五石七斗，和買本色一十一匹二丈二尺九寸四分八釐七毫五絲，折帛二十七匹二寸一分三釐七毫五絲，紬絹折帛八匹三丈九寸七寸三分九釐，紬絹本色二十匹二丈八尺九寸三分九釐。已將田四百七畝出與乙，五百十六畝出與丙。訖乞移割本户所出田上稅賦，歸併乙丙兩户送納。會到乙原有田三百七十五畝，丙原有田四百六十三畝，並係本鄉本等。每畝苗三升五合，稅一尺一寸五分，物力一貫二百；本等田紬一尺三寸四分，物力九百；物力及三十二貫數和買一匹，內三分本色七分折帛；夏稅七分本色三分折帛；紬五分本色五分折帛，併絹紬。欲知甲田地及甲乙丙分割合納苗米、和買、夏稅、折帛、本色、畸零各幾何?
［按］此題科稅分田地二項。田有科米、稅絹、物力三色，地只有稅紬、物力二色，絹與紬折色不同，物力和買合田地總計之。又甲户有田者有地，乙丙二户有田無地。此等皆不可不明言者，題中殊未分晰。
答曰：甲原有田一千二十畝，地一十一畝二角四十八步。
甲今有田九十七畝，苗米三石三斗九升五合，夏稅折帛三丈三尺四寸六分五釐，本色一匹畸零三丈八尺八分五釐，物力一百一十六貫四百文。地一十一畝二角四十八步，稅紬折帛七尺八寸三分九釐，稅紬畸零七尺八寸三分九釐，物力一十貫五百三十文，田地共物力一百二十六貫九百三十文。和買折帛二匹三丈一尺六分三釐七毫五絲，本色一疋畸零七尺五寸九分八釐七毫五絲。
乙今有田七百八十二畝，苗米二十七石三斗七升，夏稅折帛六匹二丈九尺七寸九分，本色一十五匹畸零二丈九尺五寸一分，物力九百三十八貫四百文，和買折帛二十匹二丈一尺一寸，本色八匹畸零三丈一尺九寸。
丙今有田九百七十九畝，苗米三十四石二斗六升五合，夏稅折帛八匹一丈七尺七寸五分五釐，本色一十九匹畸零二丈八尺九分五釐，物力一千一百七十四貫八百文，和買折帛二十五匹二丈七尺九寸五分，本色一十一匹畸零五寸五分。
術曰：以粟米及衰分求之。置甲原納米為實，以每畝苗為法除之，得甲原有田。以乘每畝稅，得田絹。次以甲紬絹本色折帛併之，內減田絹，餘為地紬。以每紬約之，得甲原有地。次以甲出田併乙丙原有田，各得三户今有田地。各以等則乘之，各得物力苗稅。次以每畝物力率約各户共物力，得和買。副之，三分因之退位為和本，七分因之退位為和折；夏稅反其分而因之退之；紬以半之，併入稅，各得。

均科綿稅

問：縣科綿有五等户，共一萬一千三十三户，共科綿八萬八千三百三十七兩六錢。上等一十二户，副等八十七户，中等四百六十四户，次等二千三十五户，下等八千四百三十五户。欲令上三等折半差，下二等此中等六四折差。科率求之，問各户納及各等幾何?
［按］題言下二等比中等六四折差，是次等為中等六分之四，下等又為次等六分之四也。乃術草中皆以十分之四收之，是四分折差而非四六折差矣。集中題與術不相應者多類此，豈成書之後未能重加校正歟?
答曰：上等一户一百二十四兩，一十二户計一千四百八十八兩；副等一户六十二兩，八十七户計五千三百九十四兩；中等一户三十一兩，四百六十四户計一萬四千三百八十四兩；次等一户一十二兩四錢，二千三十五户計二萬五千二百三十四兩；下等一户四兩九錢六分，八千四百三十五户計四萬一千八百三十七兩六錢。
術曰：列五等户數，先以四折下等數，加次等户，又以四折之，加中等户數，却以半折之，加二等户，又以半折之，加上等户數，不折便為法。除科綿，得上等一户之綿。復半之為副等，又半之為中等，又四折之為次等，又四折之為下等。各一户綿，却各以户數乘之，各得五等共出綿。
草曰：先置下等户八千四百三十五，以四折之，得三千三百七十四。乃以得數折入次户二千三十五內，得五千四百九户訖。又以四折次數五千四百九户，得二千一百六十三户六分。乃以得次數併入中户四百六十四內，共得二千六百二十七户六分。次以五分折中數二千六百二十七户六分，得數。次以得數一千三百一十三户八分併副户八十七，得一千四百户八分。又以五折副

數一千四百户八分，得七百户四分，既得數。乃以副數七百户四分併上户一十二户，得七百一十二户四分，不折便為法。
累折至等，共得七百一十二户四分以為總法。乃置科綿八萬八千三百三十七兩六錢為實，法七百一十二户四分而一，得一百二十四兩為上等一户所出綿。以五折之，得六十二兩為副等一户所出綿。又五折之，得三十一兩為中等一户所出綿。乃以四折之，得一十二兩四錢為次等一户所出綿。又以四折之，得四兩九錢六分為下等一户所出綿。乃以各等户數各乘一户所出，即各得每等共數之綿。

均定勸分

問：欲勸糶賑濟，據甲民户物力畝步排定，共計一百六十二户，作九等：上等三户，第二等五户，第三等七户，第四等八户，第五等十三户，第六等二十一户，第七等二十六户，第八等三十四户，第九等四十五户。今先覲諭，第一等上户願糶五千石，第九等户願糶二百石。欲知各等抛差石數併數認米數各幾何?
答曰：認該米二十三萬七千六百石。
上等一户米五千石，三户計一萬五千石；二等一户米四千四百石，五户計二萬二千石；三等一户米三千八百石，七户計二萬六千六百石；四等一户米三千二百石，八户計二萬五千六百石；五等一户米二千六百石，一十三户計三萬三千八百石；六等一户米二千石，二十一户計四萬二千石；七等一户米一千四百石，二十六户計三萬六千四百石；八等一户米八百石，三十四户計二萬七千二百石；九等一户米二百石，四十五户計九千石。
術曰：以衰分求之。置上下户米，減餘為實。列等數減一，餘為法。除之，得抛差石數。以差累減上等米，各得諸等米。以各等户乘之，併之為總數。
草曰：置上等户米五千石，減下等户二百石，餘四千八百石為實。以九等減一，餘八為法。除實，得六百石為每等抛差。用減上等，餘四千四百石為二等米。又減六百，得三千八百石為三等米。又減六百，得三千二百石為四等米。又減六百，得二千六百石為五等米。又減六百，得二千石為六等米。又減六百，得一千四百石為七等米。又減六百，得八百石為八等米。又減六百，得二百石為九等米。又各乘户數，併之，得總認米石數二十三萬七千六百石。

均定合解

問：州郡合解諸司窠名錢：户部九十六萬五千四百二十一貫文，總所六十四萬三千六百一十四貫文，運司一萬六千九十貫三百五十文。今諸窠名先催到九千二百五十三貫六百二十文，欲照原額分數均定樁收解，合各幾何?
答曰：户部五千四百九十七貫二百文；總所三千六百六十四貫八百文；運司九十一貫六百二十文。
術曰：以衰分求之。置諸原率，可約約之。副併為法。以催到錢乘未併者，各為實。實如法而一。
草曰：列户部九十六萬五千四百二十一貫，總所六十四萬三千六百一十四貫，運司一萬六千九十貫三百五十文，各為原率。今原率可約，求等得一萬六千九十貫三百五十為等數，俱約之：户部得六十，總所得四十，運司得一，各為率。副併得一百一為法。以催到錢九千二百五十三貫六百二十文乘未併數：六十、四十及一，畢得五十五萬五千二百一十七貫二百文為户部之實，三十七萬一百四十四貫八百文為總所之實，九千二百五十三貫六百二十文為運司之實。三實皆如法一百一而一：其户部得五千四百九十七貫二百文，總所得三千六百六十四貫八百文，運司得九十一貫六百二十，各為候解錢。

寬減屯租

問： 屯租欲議寬減，仍聽以夏麥折納分數。官牛種者與減二分，私牛種者與減四分。每歲租穀以三分之一許夏折二麥，內四分大六分小折色。每大麥三石折小麥二石，小麥二石折穀三石五斗。屯租舊額官種一石納租五石，私種一石納租三石。今某州屯田去年計官私種共九千七百八十二石，共合收租穀三萬九千五百八十六石。欲知官私種各數、原額、今減、合催、成年夏麥秋穀幾何?
［按］此題有官私共種數、租數，有種一石租數，求各種數租數，古謂之差分，今謂之和較比例。折色亦差分也，今謂之和數比例。大小麥與穀相易，古謂之異乘同乘，今謂之和率比列。
答曰： 官種五千一百二十石，私出種四千六百六十二石。
原租額共三萬九千五百八十六石：官種二萬五千六百石，私種一萬三千九百八十六石。
今減一萬七百一十四石四斗：官種五千一百二十石，私種五千五百九十四石四斗。
合催二萬八千八百七十一石六斗。
官種租二萬四百八十石：一分折麥計穀六千八百二十六石六斗六升零三分升之二；四分折大麥二千三百四十石五斗七升七分升之一（係折穀二千七百三十石六斗六升三分升之二）；六分折小麥二千三百四十石五斗七升七分升之一（係折穀四千九十六石二分）；正色穀一萬三千六百五十三石三斗三升三分升之一。
私種租八千三百九十一石六斗：一分折麥計穀二千七百九十七石二斗；四分折大麥九百五十九石四升（係折穀一千一百一十八石八斗八升）；六分折小麥九百五十九石四升（係折穀一千六百七十八石三斗二升）；二分正色穀五千五百九十四石四斗。
成年共收：夏折穀九千六百二十三石八斗六升三分升之二；大麥三千二百九十九石六斗一升七分升之一；小麥三千二百九十九石六斗一升七分升之一；秋正穀一萬九千二百四十七石七斗三升三分升之一。

戶稅均寬

問： 州郡寬恤，近將某縣下三等稅户秋科餘欠錢米，已與蠲放：共錢一千三百五十五貫七百六文，米五千二百七十二石一斗九升。某本縣下三等物力計三萬七千六百五十八貫五百文。今來官員陳述：本縣多有樂輸無欠之户，今蒙蠲放稅尾，似反寬潤頑輸之户，於理未均。遂議將樂輸三等户於明年兩稅與照昨來體例減免。契勘得三等無欠户物力二十二萬八千一十五貫三百二十一文。欲知每百文合減免錢米及共減各幾何?
答曰： 每物力一百文，放錢三文六分，放米一升四合。明年兩稅放免：錢七千九百四十九貫三百五十一文五分五釐六毫；米三萬九百一十四石一斗四升四合九勺四抄。
術曰： 以粟米衰分求之。置原物力為法，除原放錢米，得每百文物力所放錢米率。以各率乘今來物力，各得錢米，為明年兩稅合放數。
草曰： 以原物力三萬七千六百五十八貫五百文為法。先除原放錢一千三百五十五貫七百六文，定法百文上得一文為商，除得三文六分為物力百文所放錢率。次除原放米五千二百七十二石一斗九升，得一升四合為物力百文所放米率。以今三等户物力二十二萬八千一十五貫三百二十一文徧乘所放錢米二率，得錢七千九百四十九貫三百五十一文五分五釐六毫，得米三萬九百一十四石一斗四升四合九勺四抄，為明年兩稅合放數。

米穀粒分

問： 開倉受約，有甲户米一千五百三十四石。到廊驗得米內夾穀，乃於樣內取米一捻，數計二百五十四粒，內有穀二十八顆。凡粒米率每勺三百。今欲知米內雜穀多少，以折米數科責及粒各幾何?
答曰： 米一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八；穀一百六十九石一斗二合三勺一百二十七分勺之七十九。合折米八十四石五斗五升一合一勺一百二十七分勺之一百三。原米折米共計四十三億四千八百三十四萬六千四百五十六粒。
術曰： 以粟米求之，衰分入之。置樣米粒數為法，以常穀顆數減之，餘與穀為列衰。可約約之。以共米乘列衰為各實，實如法而一，各得米數穀數。置穀數以粟率折之，為穀所折米。次以勺率

遍乘米數折米，得粒數。
草曰：置一捻樣粒數二百五十四為法，以帶穀二十八顆為穀衰，以減法，餘二百二十六為米衰。此二衰與法皆可約，求等得二，俱以約之：法得一百二十七，米衰得一百一十三，穀衰得一十四。以米一千五百三十四石遍乘二衰，得一十七萬三千三百四十二石為米實，得二萬一千四百七十六石為穀實。皆如法一百二十七而一：米得一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八，穀得一百六十九石一斗二合三勺一百二十七分勺之七十九。以粟率五十折之，得八十四石五斗五升一合一勺一百二十七分勺之一百三為穀折納米數。併二米，得一千四百四十九石四斗四升八合八勺一百二十七分勺之二十四。先通分納子一十八萬四千八十石，以勺率三百粒乘之，得五千五百二十二億四千萬粒為實，以母一百二十七除之，得四十三億四千八百三十四萬六千四百五十六粒，不盡八十八棄之。

卷六上錢穀

課糴

問：差人五路和糴。據浙西平江府石價三十五貫文，一百三十五合，至鎮江水脚錢每石九百文；安吉州石價二十九貫五百文，一百一十合，至鎮江水脚錢每石一貫二百文；江西隆興府石價二十八貫一百文，一百一十五合，至建康水脚錢每石一貫七百文；吉州石價二十五貫八百五十文，一百二十合，至建康水脚錢每石二貫九百文；湖南潭州石價二十七貫三百文，一百一十八合，至鄂州水脚錢每石一貫七百文。其錢並十七界官會，其米並用文思院斛交量紐數。欲皆以官解計石錢相比貴賤幾何？
［按］按草中係二貫一百文，此訛。文思院解每斗八十三合。
答曰：文思院斛石錢：安吉州二十三貫一百六十四文一十一分文之六；平江府二十二貫七十一文二十七分文之二十三；隆興府二十一貫五百七文二十三分文之一十九；潭州二十貫六百七十九文五十九分文之四十九；吉州一十九貫八百八十五文十二分文之五。
［按］按三十九訛四十九。
術曰：以粟米互換求之。置石價併水脚，乘石數，又乘官斗合數為實。各如本州合數而一，各得官解石錢。以課貴賤。
草曰：置安吉州石價二十九貫五百文，平江石價三十五貫文，隆興石價二十八貫一百文，吉州石價二十五貫八百五十文，潭州石價二十七貫三百文，列右行。次置水脚：安吉一貫二百文，平江九百文，隆興一貫七百文，吉州二貫九百文，潭州二貫一百文，列左行，各對本州石價。以兩行數併之，得數：安吉三十貫七百文，平江三十五貫九百，隆興二十九貫八百，潭州二十九貫四百，吉州二十八貫七百五十，仍於右行。次以文思院官斗八十三合遍乘之：安吉州得二千五百四十八貫一百文，平江府得二千九百七十九貫七百文，江西隆興得二千四百七十三貫四百文，湖南潭州得二千四百四十貫二百文，江南吉州得二千三百八十六貫二百五十文，各為實於右。得次列安吉斗一百一十合，平江斗一百三十五合，隆興斗一百一十五合，潭州斗一百一十八合，吉州斗一百二十合於左行為法。以对除右行之實：安吉得二十三貫一百六十四文一十一分之六，平江得二十三貫七十一文二十七分文之二十三，隆興得二十一貫五百七文二十三分文之一十九，潭州得二十貫六百七十九文五十九文分之四十九，吉州得一十九貫八百八十五文一十二分文之五。相課石價，其安吉州最貴，平江次之，隆興又次之，潭州又次之，吉州最賤。

折解輕齎

問：有甲乙丙丁四郡，各合起上供銀絹。
甲郡：銀三千二百兩，每兩二貫二百文足；絹六萬四千匹，每匹二貫文足。去京一千里，每擔一里傭錢六文足。其時舊會每貫五十四文足。
乙郡：銀二千七百兩，每兩二貫三百文足；絹四萬九千二百匹，每匹二貫四百二十文足。去京九百八十里，每擔一里傭錢四文二分足。舊會價五十九文足。
丙郡：銀四千兩，每兩新會九貫三百文；絹七萬三千六百匹，每匹新會一十貫三百文。去京二千里，每擔一里傭錢八十文舊會。
丁郡：銀二千六百兩，每兩五十一貫文舊會；絹三萬二千三十五匹，每匹五十八貫文舊會。去京一千五百里，每擔一里傭錢一百文舊會。
諸郡銀每五百兩、絹每六十匹、新會每五千貫為擔。欲並折新會，均作三限起解。求各郡每限及本色原理、折解實用、寬餘、傭錢各新會幾何?
［按］此題貫數分三項：其一足數，每千文為一貫；其一舊會數，如甲以五十四文為一貫，乙以五十九文為一貫是也；其一新會數，為舊會數之五倍，如甲以二百七十文為一貫，乙以二百九十五文為一貫是也。四郡或言足數，或言舊會數、新會數。並折新會數。傭錢原以銀五百兩或絹六十匹各為一擔，今皆折新會數，以五千貫為一擔，故有寬餘錢數。題語多未詳，而併為一擔句尤混，略為分析而術草之意大概可見矣。
答曰：甲郡：合解五十萬一百四十八貫一百四十八文。初限一十六萬六千七百一十六貫四十九文，次限一十六萬六千七百一十六貫四十九文，末限一十六萬六千七百一十六貫五十文。傭錢原理二萬三千八百四十五貫九百二十五文二十七分文之二十五，實用二千二百二十二貫八百七十七文二十七分文之二十一，寬餘二萬一千六百二十三貫四十八文二十七分文之四。
乙郡：合解四十二萬四千六百五十七貫六百二十七文。初限一十四萬一千五百五十二貫五百四十二文，次限一十四萬一千五百五十二貫五百四十二文，末限一十四萬一千五百五十二貫五百四十三文。傭錢原理一萬一千五百一十六貫四百二十八文五十九分文之二十八，實用一千一百八十五貫一十文二百九十五分文之五，寬餘一萬三百三十一貫四百一十八文五十九分文之二十七。
丙郡：合解七十九萬五千二百八十貫文。初限二十六萬五千九十三貫三百三十三文，次限二十六萬五千九十三貫三百三十三文，末限二十六萬五千九十三貫三百三十四文。傭錢原理三萬九千五百九貫三百三十三文三分文之一，實用五千八十九貫七百九十二文，寬餘三萬四千四百一十九貫五百四十一文三分文之一。
丁郡：合解三十九萬八千一百二十六貫文。初限一十三萬二千七百八貫六百六十六文，次限一十三萬二千七百八貫六百六十六文，末限一十三萬二千七百八貫六百六十八文。傭錢原理一萬四千一百七十三貫五百文，實用二千三百八十八貫七百五十六文，寬餘一萬一千七百八十四貫七百四十四文。
術曰：以均輸求之。置各郡銀絹，乘各價，併之，歸足原，展足為舊會，次以五約舊會為新會，各得合解錢。以限數除之，得每限錢，不盡併歸末限。次置里數，乘每里傭價為率。以率乘原銀及原絹，各為傭實。以每擔銀絹率各為法，實如法而一，不滿者亦為擔，併之為原理傭錢。次以率乘合觧錢為實，乃以錢物每擔率為法，實如法而一，各得實用傭錢。以減原理傭錢，餘為寬餘傭錢。

卷六下錢穀

僦直推原

問：房廊數內，一户日納一百五十六文八分足為準。指揮未曾經減者，減三分；已曾經減三分者，減二分；已曾經減二分者，更減二分。今本户累經減者，欲知原額房錢幾何？
答曰：原額三百五十文。
術曰：以衰分求之。列一十分兩行，各三位；列減分，對減右行。以餘者相乘為法。以左行原列相乘，得納錢為實。實如法而一，得原額錢。
草曰：列一十三位於左行，又列一十分三位於右行。其右上減去初減三分，右中減去次減二分，右下減去更減二分。右行餘七八八，以相乘得四百四十八為法。乃以左行三位一十分相乘，得一千為乘率。以乘見日納錢一百五十六文八分，得一百五十六貫八百文為實。實如法而一，得三百五十文，為本户原額戸錢。

推求本息

問：三庫息例：萬貫以上一釐，千貫以上二釐五毫，百貫以上三釐。甲庫本四十九萬三千八百貫，乙庫本三十七萬三百貫，丙庫本二十四萬六千八百貫。今三庫共約到息錢二萬五千六百四十四貫二百文。其典率：甲反錐差，乙方錐差，丙蒺藜差。欲知原典三例本息各幾何？
[按] 此即衰分題也。其差有反錐、方錐、蒺藜之名，蓋以一二三遞減如立錐為反錐，以一四九平方遞加為方錐，以一三六三數遞加為蒺藜。是必古有其名也。至以各差求各本，則因各本原依各差入之也。
答曰：甲庫共納息九千五十三貫文：一釐息二千四百六十九貫文，二釐半息四千一百一十五貫文，三釐息二千四百六十九貫文。
乙庫共納息一萬五十一貫文：一釐息二百六十四貫五百文，二釐半息二千六百四十五貫文，三釐息七千一百四十一貫五百文。
丙庫共納息六千五百四十貫二百文：一釐息二百四十六貫八百文，二釐半息一千八百五十一貫文，三釐息四千四百四十二貫四百文。
術曰：置諸庫諸色之差，照釐率為三行。縱併之為約率，横命之為乘率。以約率各約自庫之本，各得。以遍乘未併乘率，然後各以釐率横乘之，次以縱併之，為各庫共息。
草曰：置甲庫反錐差，自下置三二一於右行。次置乙庫方錐差，自上置一四九於中行。次置丙庫蒺藜差，自上置一三六於左行。各為三庫上中下三等乘率。乃縱併甲差三二一，得六為甲約率。縱併乙差一四九，得一十四為乙約率。縱併丙差一三六，得一十為丙約率。直命九位數各為上中下乘率。乃先以約率各約自庫之本：乃以甲約率六約甲本四十九萬三千八百貫，得八萬二千三百貫為甲得。次以乙約率一十四約乙本三十七萬三百貫，得二萬六千四百五十貫為乙得。次以丙約率一十約丙本二十四萬六千八百貫，得二萬四千六百八十貫為丙得。以各得乘未併乘率：其甲所得八萬二千三百貫，乘反錐乘率三二一，得二十四萬六千九百貫為上率，得一十六萬四千六百貫為中率，得八萬二千三百貫為下率。其乙所得二萬六千四百五十貫，以乘方錐差一四九，得二萬六千四百五十貫為上率，得一十萬五千八百貫為中率，得二十三萬八千五十貫為下率。其丙所得二萬四千六百八十貫，以乘蒺藜差一三六，得二萬四千六百八十貫為上率，得七萬四千四十貫為中率，得一十四萬八千八十貫為下率。然後各以息釐數乘各庫三乘：其甲以一釐乘上率二十四萬六千九百貫，得二千四百六十九貫為上息；以二釐五毫乘中率一十六萬四千六百貫，得四千一百一十五貫為中息；以三釐乘下率八萬二千三百貫，得二千四百六十九貫為下息。併上中下三息，得九千五十三貫文為甲庫共息。其乙庫以一釐乘上率二萬六千四百五十貫，得二百六十四貫五百文為上息；以二釐五毫乘中率一十萬五千八百貫，得二千六百四十五貫為中息；以三釐乘下率二十三萬八千五十貫，得七千一百四十一貫五百為下息。併上中下三息，得一萬五十一貫文為乙庫共息。其丙庫以一釐乘上率二萬四千六百八十貫，得二百四十六貫八百為上息；以二釐五毫乘中率七萬四千四十貫，得一千八百五十一貫為中息；以三釐乘下率一十四萬八千八十貫，得四千四百四十二貫四百文為下息。併上中下三息，得六千五百四十貫二百文為丙庫共息。併三庫共息，得二萬五千六百四十四貫二百文為總息。

易牒知原

［按］按舊本此問無題，今增入。
問：出度牒差人營運：每三道易鹽一十三袋，鹽二袋易布八十四疋，布一十五疋易絹三疋半，絹六疋易銀七兩二錢。今趁到銀九千一百七十二兩八錢。欲知原闕度牒道數幾何？
答曰：度牒一百八十道。
術曰：以粟米互乘易法求之。列各數以本色相對，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。
草曰：先以度牒三道乘鹽二袋，得六。以乘布一十五，得九十。又乘絹六疋，得五百四十。乃乘銀九萬一千七百二十八錢，得四千九百五十三萬三千一百二十錢為實。次以鹽一十三袋乘布八十四，得一千九十二。以乘絹三疋五分，得三千八百二十二。乃乘銀七兩二錢，得二十七萬五千一百八十四錢為法。除實，得一百八十道為原闕度牒。

粟米交易

［按］按舊本此問無題，今增。
問：菽三升易小麥二升，小麥一升五合易油麻八合，油麻一升二合易粳米一升八合。今將菽一十四石四斗欲易油麻，又將小麥二十一石六斗欲易粳米。問各幾何？
答曰：油麻五石一斗二升；粳米一十七石二斗八升。
術曰：以粟米換易求之。置原易率，本色對列，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。各得或問數，不干其率者不置。
草曰：置四色六數，列六位率，如鴈翅，皆化為合。先將菽一十四石四斗化作一萬四千四百合，乃對前二句率數四位，如鴈翅，至欲易油麻止，共五事為上圖。次將小麥二十一石六斗化為二萬一千六百合，乃對後兩句率四位，鴈翅，至欲粳米止，共五事為下圖。
下圖：其上圖以菽一萬四千四百合乘麥二十，得二十八萬八千，又乘油麻八合，得二百三十萬四千合為油麻實。次以菽三十合乘麥一十五合，得四百五十合為法。除之，得五千一百二十合，展為五石一斗二升為油麻。
其下圖以小麥二萬一千六百合乘油麻八合，得一十七萬二千八百合，又乘粳米一十八合，得三百一十一萬四百合為粳米實。以小麥一升五合乘油麻一十二合，得一百八十合為法。除之，得一萬七千二百八十，展作一十七石二斗八升為粳米。

計米易麴

［按］按舊本此問無題，今增。
問：庫率：粳穀七石出米三石，糯米一斗易小麥一斗七升，小麥五升踏麴二斤四兩，麴一百一十斤醞糯米一石三斗。今有糯穀一千七百五十九石三斗八升，欲出穀做米，易麥踏麴，還自醞，餘穀之米須令適足。各合幾何？
答曰：共穀一千七百五十九石三斗八升。出穀九百二十四石，得米三百九十六石。易麥六百七十三石二斗。踏麴三萬二百九十四斤。餘穀八百三十五石三斗八升。醞米三百五十八石二升。
術曰：以粟米換易求之。置諸率，隨本色對列，如鴈翅。有分者通之，異類者變之，以頭位者進乘之，以下位退乘之，得合數。有對者相乘之，無對者直命之，為諸率。併上下無對者為法率。以今有物徧乘諸率（不乘法率），各為實。諸實並如法而一，各得其已變者，復互易乘除之，即得所求。
草曰：置糯穀七、出米三於右行上副兩位。次置糯米一斗、麥一斗七升於副行副中兩位。次置小麥五升、踏麥二斤四兩於次行中次兩位。次置麴一百一十觔、醞米一石三斗於左行次下兩位，隨本色對列，如鴈翅。訖乃驗次行二斤四兩是四分斤之一，以母四通次行兩位，以子一內次行次位，具中位得二，次位得九。又驗左行下位是糯米，是異類，於糯米合變為糯穀，乃以問中首句率穀七米三變之，以七因米一石三斗得九石一斗於左下為穀，却以米三因麴一百一十斤為三百三十斤麴於左行，得變圖數。以左行三百三十乘次行二，得六百六十。次以六百六十乘副行一十，得六千六百。次以六千六百乘右上七，得四萬六千二百，各於原位。却以右行副位三因副行一斗七升，得五斗一升。又以五斗一升乘次行九，得四百五十九。又以四百五十九乘左下九

十一，得四萬一千七百六十九，列為合圖數。乃驗合圖四行，其副中次三位有對，以對相乘合之。其右上左下無對者，直命之，皆為率。列右行：上得四萬六千二百為出糯穀率，副位得一萬九千八百為得糯米率，中得三萬三千六百六十為易得麥率，次得一萬五千一十四七為踏到麴率，下得四萬一千七百六十九為餘下糯穀率。併上下率，共得八萬七千九百六十九為法率。今六率共求等得一，約之，只得原率為率圖。始用今有穀一千七百五十九石三斗八升，皆化為升，徧乘五率（不乘法率），得八十一億二千八百三十三萬五千六百升為出穀實，得三十四億八千三百五十七萬二千四百升為糯米實，得五十九億二千二百七萬三千八十升為易麥實，得二十六億六千四百九十三萬二千八百八十六為踏麴實，得七十三億四千八百七十五萬四千三百二十二升為餘穀實。其五實皆如法八萬七千九百六十九而一：得九百二十四石為出穀，得三百九十六石為做到糯米，得六百七十三石二斗為易到小麥，得三萬二百九十四斤為踏到麴，得八百三十五石三斗八升為餘下穀。今將餘下穀變為米，乃以乘率三因餘穀八百三十五石三斗八升，得二千五百六石一斗四升為實，以糯穀率七為法除之，得三百五十八石二升為醞米。

囘運費

問： 有江西水運米一十二萬三千四百石，原係鎮江交卸，計水程二千一百三十里，每石水脚錢一貫二百文。今截上件米就池州安頓，池州至鎮江八百八十里。欲收囘不該水脚錢幾何?
答曰： 收囘錢六萬一千一百七十八貫五百九十一文。
術曰： 以粟米互易求之。置池州至鎮江里數，乘水脚錢，得數又乘運米為實。以原至鎮江水程為法，除實，得收囘錢。
草曰： 置池州至鎮江八百八十里，乘每石水脚錢一貫二百，得一千五十六貫文。又乘運米一十二萬三千四百石，得一億三千三十一萬四百貫文為實。以原至鎮江水程二千一百三十里為法，除實，得六萬一千一百七十八貫五百九十一文為收囘錢。

三色均價

問： 庫有三色金共五千兩：內八分金一千二百五十兩，兩價四百貫文；七分五釐金一千六百兩，兩價三百七十五貫文；八分五釐金二千一百五十兩，兩價四百二十五貫文。並欲煉為足色，每兩工食藥炭錢三貫文，耗金九百七十二兩五錢。欲知色分及兩價各幾何?
答曰： 色十分；兩價五百三貫七百二十四文，五百三十七分文之二百一十二。
術曰： 以方田及粟米求之。置共數，以耗減之，餘為法。以三色分數各乘兩數，併之為色分實。以三色價數各乘兩數為寄，以工藥價乘共金，併寄，共為價實。二實皆如法而一，即各得。
草曰： 置共金五千兩，減耗九百七十二兩五錢，外餘四千二十七兩五錢為法。次置一千二百五十兩，乘八分，得一萬分於上。置一千六百兩，以七分五釐乘之，得一萬二千分，加上。置二千一百五十兩，乘八分五釐，得一萬八千二百七十五分，又加上，共得四萬二百七十五分為分實。次置一千二百五十兩乘四百貫，得五十萬貫為寄。次置一千六百兩乘價三百七十五貫，得六十萬貫，加寄。次置二千一百五十兩乘價四百二十五貫，得九十一萬三千七百五十貫，又加寄。次置共金五千兩，乘工藥錢三貫，得一萬五千貫，又加寄，共得二百二萬八千七百五十貫為貫實。二實並如法四千二十七兩五錢而一：得其色一十分；其價每兩得五百三貫七百二十四文，不盡一貫五百九十。與法求等，得七十五，俱約之，為五百三十七文之二百一十二。

數學九章賦役錢穀四庫全書

問

答

術

草

道古

大衍

天時

田域

測望

賦役

錢穀

營建

軍旅

市易

[按]

數學九章

卷七至卷九

作者 （宋）秦九韶撰

分類 營建類 · 軍旅類 · 市易類

四庫全書

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卷七上

營建類

計作清臺問： 問

□築清臺一所，正高一十二丈，上廣五丈，袤七丈，下廣一十五丈，袤一十七丈。其袤當東西，廣當南北。北秋程人日行六十里，里法三百六十步。□土鍬土每工各二百尺，築土每工九十尺，每擔土壤一丈遠。今欲築此臺，問：需要多少工日？

答曰： 開

方及積尺數，以定工日。用工若干萬工。

術曰： 按

臺體□塹堵，用上廣袤、下廣袤及高相求，得積數。臺積術曰：上下廣袤各自相乘，又上下廣袤交叉相乘，□之，以高乘之，三而一，得臺積。以各工率除之，得工日。

塹堵

草曰： 上

廣$=5$丈，上袤$=7$丈，下廣$=15$丈，下袤$=17$丈，高$=12$丈。

$$上廣\times上袤 = 5 \times 7 = 35$$

$$下廣\times下袤 = 15 \times 17 = 255$$

$$上廣\times下袤 = 5 \times 17 = 85$$

$$上袤\times下廣 = 7 \times 15 = 105$$

四數相并：$35 + 255 + 85 + 105 = 480$

以高乘之：$480 \times 12 = 5760$

三而一：$\dfrac{5760}{3} = 1920$（立方丈）

□□立方尺：$1920 \times 1000 = 1{,}920{,}000$（立方尺）

按各工率均攤，得總工若干。

計築□岸問： 問

築□一道，長一千八百丈。其□□截面：上闊一丈，下闊三丈，高一丈二尺。問：積土若干？用夫若干？

答曰： 積

土二百八十八萬立方尺，用夫若干。

術曰： □

體□長條截頭體。以上下闊并之，半之，得平均闊；以高乘之，得截面積；以長乘之，得積。術曰：上下廣各一，并而半之，以高乘之，又以長乘之，得堤積。

草曰： 上

闊$=1$丈，下闊$=3$丈，高$=1.2$丈，長$=1800$丈。

上下闊并半：$\dfrac{1+3}{2} = 2$（平均闊）

截面積：$2 \times 1.2 = 2.4$（平方丈）

積土：$2.4 \times 1800 = 4320$（立方丈）

□尺：$4320 \times 1000 = 4{,}320{,}000$（立方尺）

計開河渠問： 問

開河渠一道，長三千六百丈。渠口上廣八丈，底廣四丈，深二丈。每工日穿土六十尺，負土一尺遠。問：積土及用工各若干?

答曰： 積

土若干立方尺，用工若干萬工。

術曰： 渠

體🄵塹堵。上下廣并之，半之，以深乘之，得截面積；又以長乘之，得積。術同堤法。

草曰： 上

廣$=8$丈，底廣$=4$丈，深$=2$丈，長$=3600$丈。

上下廣并半：$\dfrac{8+4}{2}=6$（平均闊）

截面積：$6\times 2=12$（平方丈）

積土：$12\times 3600=43200$（立方丈）

🄵尺：$43200\times 1000=43{,}200{,}000$（立方尺）

用工：$\dfrac{43200000}{60}=720{,}000$（工日）

卷七下

營建類（續）

計作石坝問： 問

作石坝一座，長五十丈，高一十丈，迎水面斜長十二丈，背水面斜長八丈。頂闊三丈，底闊一十丈。問：積石若干？

答曰： 積

石若干立方丈。

術曰： 坝

體截面F梯形。以頂闊、底闊并之，半之，以高乘之，得截面積；以長乘之，得積。

草曰： 頂

闊$=3$丈，底闊$=10$丈，高$=10$丈，長$=50$丈。

上下并半：$\dfrac{3+10}{2}=6.5$

截面積：$6.5\times10=65$（平方丈）

積石：$65\times50=3250$（立方丈）

計修城池問： 問

修城一段，城周回一千二百丈。其城截面：頂闊二丈五尺，底闊八丈，高三丈。又有女F二十四座，每座長二丈，闊一丈，高八尺。問：共積土若干？

女F

答曰： 城

積及女F積共若干立方丈。

術曰： 城

體F環形截頭體，以周回F長。術曰：頂底闊并半，以高乘之，得截面積；以城周回乘之，得城積。女F各F長方體，長寬高相乘得積，以座數乘之。并城F與女F積，得總數。

草曰： 城

截面：頂闊$=2.5$丈，底闊$=8$丈，高$=3$丈，周$=1200$丈。

上下并半：$\dfrac{2.5+8}{2}=5.25$

城截面積：$5.25\times3=15.75$（平方丈）

城積：$15.75\times1200=18900$（立方丈）

女F積：$2\times1\times0.8=1.6$（立方丈座）

女F總積：$1.6\times24=38.4$（立方丈）

總積：$18900+38.4=18938.4$（立方丈）

計造浮橋問： 問

造浮橋一所以濟師。河闊三百丈，每舟長五丈，闊一丈二尺，深八尺。舟上鋪板，板厚六寸。問：需舟若干？板積若干？

答曰： 需

舟若干艘，板積若干立方丈。

術曰： 以

河闊除以舟長，得舟數。以舟面積乘板厚，得板積。

草曰： 河

闊$=300$丈，舟長$=5$丈，舟闊$=1.2$丈。

舟數：$\dfrac{300}{5} = 60$（艘）
每舟面積：$5 \times 1.2 = 6$（平方丈）
板積：$6 \times 0.06 \times 60 = 21.6$（立方丈）

計開倉窖問： 問
開倉窖一所，上口方二丈，底方一丈二尺，深一丈八尺。問：容積及積土各若干?

答曰： 容
積若干立方丈，積土若干立方丈。
術曰： 倉
窖🄵方錐臺。術曰：上下方各自乘，上下方相乘，并之，以深乘之，三而一。
草曰： 上
方$= 2$丈，下方$= 1.2$丈，深$= 1.8$丈。
上方自乘：$2 \times 2 = 4$
下方自乘：$1.2 \times 1.2 = 1.44$
上下方相乘：$2 \times 1.2 = 2.4$
三數并：$4 + 1.44 + 2.4 = 7.84$
以深乘：$7.84 \times 1.8 = 14.112$
三而一：$\dfrac{14.112}{3} = 4.704$（立方丈）

卷八上

軍旅類

計立方營問： 問

一軍三將，將三十三隊，隊一百二十五人。遇暮立營，人占地方八尺。須令隊間容隊，師居中央。欲知營方幾何？

答曰： 答

曰：方營一百七十一丈，隊方九丈。

術曰： 先

計總人數，以每人占地八尺乘之，得營總積。乃開平方，得營方。又以每隊人數乘占地，開平方，得隊方。營方減隊方而半之，得隊距。

草曰： 總

隊數：$3 \times 33 = 99$（隊）

總人數：$99 \times 125 = 12375$（人）

營總積：$12375 \times 8 = 99000$（平方尺）

🄵🄵平方丈：$\frac{99000}{100} = 990$（平方丈）

開平方得營方：$\sqrt{990} \approx 31.5$（丈）按古法修正🄵整數。

按三三隊布陣，營方：171（丈），隊方：9（丈）。

計圓營問： 問

立圓營一匝，馬軍五千騎，步軍三萬人。騎兵占地二丈四尺，步兵占地九尺。令步居🄵，騎居外，各🄵方陣。問：營圓徑幾何？

答曰： 圓

營徑若干丈。

術曰： 以

人數各乘占地，得步兵積、騎兵積。各開平方，得🄵外方邊。以🄵方加兩倍騎陣厚，得外方。乃以方求圓，得營徑。

草曰： 步

兵積：$30000 \times 0.9 = 27000$（平方丈）

步兵方邊：$\sqrt{27000} \approx 164.3$（丈）

騎兵積：$5000 \times 2.4 = 12000$（平方丈）

騎兵方邊：$\sqrt{12000} \approx 109.5$（丈）

按圓徑術求之，得營圓徑若干丈。

計望敵🄵問： 問

敵軍臨河下寨。於我岸高山上，立一表高三丈。🄵望敵營，表末與敵人目及。退行一十丈，伏地望之，表末與敵人目又及。問：敵營去表及敵人各若干？敵🄵約幾何？

表

答曰： 敵

營去表若干丈，敵人目高若干尺，敵🄵約若干。

術曰： 此

🄵重差之術。以表高乘退行🄵實，以兩望之差🄵法，實如法而一，得敵營去表之遠。又以表高乘表去敵營，以退行除之，得敵人目高。以營地積及人占推之，得敵🄵約數。

卷八下

軍旅類（續）

計軍衣布問： 問

今有軍三萬二千四百人，每人給[F]一腰，用布七尺五寸；給襖一領，用布一丈二尺；給被一條，用布二丈四尺。問：共布幾何？

[F]襖被

答曰： 答

曰：共布若干匹（每匹四丈）。

術曰： 以

人數乘每人用布，得總布數。術曰：各以人數乘本用布，并而[F]總。以匹法四丈除之，得匹數。

草曰： 每

人用布：$7.5+12+24=43.5$（尺）

總布：$32400\times 43.5=1{,}409{,}400$（尺）

[F]匹：$\dfrac{1409400}{40}=35{,}235$（匹）

計造軍衣問： 問

造軍衣，每人一襲。每襲用布一丈四尺，裏布七尺，[F]一兩二錢，棉一斤八兩。今有軍一萬八千人，問：物料各若干？

答曰： 布

若干匹，裏布若干匹，[F]若干斤，棉若干斤。

術曰： 以

軍人數各乘每襲所用物料，得總數。以各法約之。

草曰： 面

布總：$18000\times 14=252{,}000$（尺）$=6300$（匹）

裏布總：$18000\times 7=126{,}000$（尺）$=3150$（匹）

[F]總：$18000\times 1.2=21{,}600$（錢）$=135$（斤）

棉總：每人棉$=1$斤8兩$=1.5$斤，$18000\times 1.5=27000$（斤）

計載糧運問： 問

運糧十萬石，每車載五十石。車行日行六十里，空車日行八十里。裝卸各一日。問：車若干輛？日若干？

答曰： 車

若干輛，往返若干日。

術曰： 以

總糧除以每車載數，得車數。以行程及空重車速，求往返日數。術曰：總糧[F]實，每車載[F]法，實如法而一，得車數。又以里數求空重日行，并裝卸日，得往返總日。

草曰： 車

數：$\dfrac{100000}{50}=2000$（輛）

裝車日：1日，卸車日：1日

重車日：按行程若干里，重車行60里日

空車日：同行程，空車行80里日

往返總日$=1+1+$重車日$+$空車日

計營糧問： 問
一軍二萬五千人，每人日食米二升。今有米三萬石，問：足幾日食?

答曰： 足
若干日食。
術曰： 以
人數乘日食量，得日食總數。以總米除以日食數，得足日。
草曰： 日
食總：$25000 \times 2 = 50000$（升）$= 500$（石）
足日：$\dfrac{30000}{500} = 60$（日）

卷九上

市易類

計🄵米粟問： 問

🄵米三千石，每石價錢二貫五百文。今持銀一錠，重五十兩，每兩折錢一貫二百文。問：銀足否？余欠若干？

答曰： 銀

不足，欠若干貫。

術曰： 以

米數乘石價，得總價。以銀重乘每兩折錢，得銀值。以銀值減總價，得余欠。

草曰： 總

價：$3000 \times 2500 = 7{,}500{,}000$（文）$= 7500$（貫）

銀值：$50 \times 1200 = 60{,}000$（文）$= 60$（貫）

余欠：$7500 - 60 = 7440$（貫）

計均貨值問： 問

甲有絹三百匹，每匹價三貫；乙有布五百匹，每匹價一貫二百文；丙有🄵二百匹，每匹價五貫。三人欲共買一船貨，價一萬貫。問：各出幾何？

絹布🄵

答曰： 甲

出若干貫，乙出若干貫，丙出若干貫。

術曰： 此

🄵均輸之術。以各人所持貨值🄵率，按率分配總價。術曰：各以匹數乘本價，得總值🄵列衰。并以🄵法，以總價乘列衰，實如法而一，各得所出。

草曰： 甲

值：$300 \times 3 = 900$（貫）

乙值：$500 \times 1.2 = 600$（貫）

丙值：$200 \times 5 = 1000$（貫）

總值：$900 + 600 + 1000 = 2500$（貫）

甲出：$\frac{900}{2500} \times 10000 = 3600$（貫）

乙出：$\frac{600}{2500} \times 10000 = 2400$（貫）

丙出：$\frac{1000}{2500} \times 10000 = 4000$（貫）

計求美茶問： 問

官買茶三處：甲處茶八千斤，每斤價三百二十文；乙處茶一萬二千斤，每斤價二百八十文；丙處茶一萬五千斤，每斤價二百五十文。欲均🄵一價發賣，問：每斤均價幾何？

答曰： 均

價若干文。

術曰： 此

🄵均價之術。以各處斤數乘本價，得各處總價；并之🄵實；并各處斤數🄵法；實如法而一，得均價。

草曰： 甲

總價：$8000 \times 320 = 2{,}560{,}000$（文）

乙總價：$12000 \times 280 = 3{,}360{,}000$（文）
丙總價：$15000 \times 250 = 3{,}750{,}000$（文）
總價：$2560000 + 3360000 + 3750000 = 9{,}670{,}000$（文）
總斤：$8000 + 12000 + 15000 = 35000$（斤）
均價：$\dfrac{9670000}{35000} \approx 276.3$（文斤）

卷九下

市易類（續）

計求鹽價問： 問
官鬻鹽，每袋重三百斤，成本一百貫。欲得利三成，問：每袋賣價幾何？每斤賣價幾何？

答曰： 每
袋賣價若干貫，每斤若干文。
術曰： 以
成本乘利加成本，得賣價。術曰：以本🄵一，加三成🄵一三，以本乘之，得袋價。以袋價除以斤數，得斤價。
草曰： 成
本$= 100$貫，利$= 30$
袋價：$100 \times 1.3 = 130$（貫）
斤價：$\dfrac{130000}{300} \approx 433.3$（文斤）

計均錢值問： 問
庫中舊錢三種：一曰足陌錢，陌法一千文；二曰九陌錢，陌法九百文；三曰八陌錢，陌法八百文。今有足陌錢五百貫，九陌錢三百貫，八陌錢二百貫。欲均🄵一陌，問：合一千文🄵陌，共得若干貫？
足陌錢九陌錢八陌錢
答曰： 共
得若干貫（足陌）。
術曰： 以
各陌錢數乘本陌法，得實際文數；并之；以足陌法一千除之，得共數。
草曰： 足
陌實數：$500 \times 1000 = 500{,}000$（文）
九陌實數：$300 \times 900 = 270{,}000$（文）
八陌實數：$200 \times 800 = 160{,}000$（文）
總實數：$500000 + 270000 + 160000 = 930{,}000$（文）
合足陌：$\dfrac{930000}{1000} = 930$（貫）

計定物價問： 問
有絲、綿、麻三物，共值八百貫。絲一斤價四貫，綿一斤價二貫，麻一斤價五百文。欲買絲、綿、麻各若干斤，使價相等。問：各得幾何？
絲綿麻
答曰： 絲
若干斤，綿若干斤，麻若干斤。
術曰： 以
總價三而一，得每物之值。各以本價除之，得斤數。
草曰： 每
物之值：$\dfrac{800}{3} \approx 266.67$（貫）
絲斤數：$\dfrac{266.67}{4} \approx 66.67$（斤）
綿斤數：$\dfrac{266.67}{2} \approx 133.33$（斤）
麻斤數：$\dfrac{266.67}{0.5} = 533.33$（斤）

計販運得利問： 問

一人持絹五百匹至遠方販賣。去時每匹價四貫，到彼時價漲五成。回程買茶，茶價每斤三百文。問：賣絹得錢若干？可買茶若干斤？

答曰： 得

錢若干貫，可買茶若干斤。

術曰： 以

本價加利，得賣價；以匹數乘之，得總錢。以總錢除以茶價，得茶斤數。

草曰： 賣

價：$4 \times 1.5 = 6$（貫匹）

總錢：$500 \times 6 = 3000$（貫）$= 3{,}000{,}000$（文）

茶斤數：$\dfrac{3000000}{300} = 10000$（斤）

丈	≈ 3.33
尺	≈ 33.3
寸	≈ 3.33
平方丈	≈ 11.09
立方丈	≈ 37.0
立方尺	≈ 37.0
石	≈ 60
升	≈ 0.6
斤	≈ 600
兩	≈ 37.5
錢	≈ 3.75
貫	
文	
里	≈ 556
步	≈ 1.54

$a \times bc \times dh$

$$V = \frac{h}{3}(ab + cd + ad + bc)$$

衰分

重差術

陌